Psychedelic medicine: mechanisms, evidence, and translation to practice

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Series explanation: State of the Art Reviews are commissioned on the basis of their relevance to academics and specialists in the US and internationally.

ABSTRACT

Over the past 15 years, psychedelic treatments have garnered substantial clinical interest, with psilocybin and 3,4-methylenedioxymethamphetamine (MDMA) advancing to phase 3 trials for various psychiatric conditions. This state-of-the-art review examines the evidence for these treatments and their proposed mechanisms of action, and identifies key challenges in clinical translation. Psychedelic treatments combine limited drug administration with psychotherapy or psychological support, potentially working through various biological and psychological pathways, including effects on neural circuitry, emotional processing, and psychological flexibility, though the precise mechanisms are not completely understood. The strongest evidence supports psilocybin for treatment resistant depression and MDMA for post-traumatic stress disorder, with emerging data in substance use disorders and psychological distress in life threatening illness. Although psychedelics are well tolerated under controlled conditions, methodological limitations complicate interpretation of the evidence, including functional unblinding, variable psychotherapeutic protocols, and homogeneous trial populations. Current treatment models require extensive therapeutic support in specialized settings, posing challenges with scalability. While innovative approaches like group administration may improve accessibility, their comparative effectiveness remains to be established. Future priorities for the research field include standardizing adverse event assessment, developing evidence based implementation guidelines, and showing durable benefit in diverse patient populations. Whether psychedelic treatments can fulfil their therapeutic promise while meeting regulatory requirements and healthcare system constraints remains a defining question for the field.

Introduction

The past 15 years have seen surging research and clinical interest in psychedelic therapies, driven by promising results with psilocybin and 3,4-methylenedioxymethamphetamine (MDMA). Both substances have shown possible applications across several conditions, including affective disorders, post-traumatic stress disorder (PTSD), and substance use disorders. Encouraging early outcomes have led to several breakthrough therapy designations by the United States Food and Drug Administration (FDA). Given limited pharmacotherapy innovation in recent decades, these preliminary findings have generated substantial, though often premature,[1] interest and excitement.

Some argue that enthusiasm for psychedelics has outpaced evidence quality. Critics cite methodological limitations, potential safety concerns, and risks of lowered regulatory standards through expedited approval.[2] The tension between clinical need and robust evidence is exemplified by the recent FDA rejection of the license application by Lykos Therapeutics for MDMA assisted therapy in PTSD, underscoring ongoing debate about trial design and evidence quality.[3 4]

Although "classic" psychedelics like psilocybin and lysergic acid diethylamide (LSD) are pharmacologically distinct from "entactogens" or "empathogens" such as MDMA, both share striking, rapid onset subjective effects and prolonged impacts on behavior and cognition well after metabolism. Both are used in a distinctive treatment model (fig 1, fig 2). Rather than daily dosing, psychedelics are administered one to three times within a sequence of preparatory, drug administration, and integration sessions to reflect on the experience (fig 1, fig 2).

These features create challenges across several domains. Methodological issues include the pronounced subjective effects that compromise study blinding, while the resource intensive treatment

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Preparation	Administration	Integration
• Multiple sessions establishing therapeutic rapport and alliance • Setting therapeutic intentions or goals • Psychoeducation about acute drug effects and potential experiences • Development of coping strategies or grounding techniques for challenging experiences • Assessment of readiness and risk factors	Common elements	• Several non-drug sessions after each administration • Processing and making meaning of experiences • Translating insights into behavioral change • Supporting psychological stability • Duration and frequency varies by protocol (typically two to six sessions)
	• Drug sessions lasting 6-8 hours • Medical monitoring and rescue drugs available • One to three sessions depending on protocol • One or more therapists present throughout • Encouragement from the therapists to be open to the experience that unfolds	
	Classic psychedelics (eg, psilocybin)	
	• Pronounced alterations in consciousness (cognition, affect, and perception) • Patient reclines with eyeshades and music throughout • Minimal dialogue; emphasis on non-directive inner exploration • Therapists maintain supportive presence primarily through silent witnessing • Brief check-ins mainly to ensure safety and comfort; active support offered when needed	
	MDMA for PTSD	
	• Primary effects on emotional processing and interpersonal trust • Less intense perceptual effects than classic psychedelics • Therapists actively support emotional processing of a trauma while following participant's lead • Dynamic balance between inner focus and therapeutic dialogue, shifting between periods of silent inner processing, verbal expression of emerging memories and emotions, and active engagement with therapists	

Fig 1 | Psychedelic assisted therapy treatment models combine pharmacological intervention with structured psychological support in three distinct phases. MDMA=3,4-methylenedioxymethamphetamine; PTSD=post-traumatic stress disorder

model contributes to small sample sizes and limits long term follow-up feasibility. Implementation faces substantial barriers stemming from the treatment model, including therapist training requirements, high per patient clinician time costs, and specialized facility requirements. Additionally, safety monitoring is complicated by variable adverse event assessment that risks systematic under-reporting despite published trials reporting high tolerability.[5]

Although the term "psychedelic" encompasses mechanistically diverse compounds including ketamine and ibogaine, the current review focuses on classic serotonergic psychedelics (principally psilocybin) and MDMA as the compounds most advanced in clinical development. These substances are examined together based on their overlapping therapeutic approach of limited drug administration within psychotherapy or structured psychological support (fig 1), rather than strict pharmacological similarity. In this review, we describe each compound, survey candidate mechanisms of action, assess the clinical evidence base, discuss translational challenges, and suggest future directions for research.

Methods

Sources and selection criteria

We identified articles by searching bibliographic databases including PubMed (January 2000 to January 2025), limited to English language publications. Box 1 outlines the inclusion and exclusion criteria applied. To ensure comprehensive coverage while maintaining focus on clinically relevant evidence, we prioritized randomized controlled trials, systematic reviews, and meta-analyses. We also included key mechanistic and translational studies that inform clinical applications. The online supplement provides a representative PubMed search strategy. Database searches were supplemented by forward citation tracking of key papers, reference list screening of recent systematic reviews, hand searching of specialized journals, and consultation with subject matter experts to identify additional relevant studies.

Mechanisms of action of classic psychedelics

The therapeutic effects of classic psychedelics involve interacting pathways across molecular, circuit, and psychological levels. Research has identified specific mechanisms from serotonin 2A ($5\text{-}HT_{2A}$) receptor activation and neuroplasticity enhancement to brain network reorganization and cognitive flexibility changes. These mechanisms may operate synergistically or contribute independently to therapeutic benefit through pluralistic causation.[6] This section reviews current evidence organized across three levels: molecular and cellular foundations, circuit and network reorganization, and cognitive and behavioral outcomes, acknowledging that therapeutic effects likely emerge from complex interactions between these levels.

Fig 2 | Schematic of various sessions involved in participant dosing studies. These components form the core of the psychedelic clinical trial paradigm

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Box 1: Literature search selection criteria

Inclusion criteria
- Peer reviewed articles
- Randomized controlled trials, meta-analyses, systematic reviews, and observational studies
- Studies focusing on psychiatric conditions (eg, depression, post-traumatic stress disorder, substance use disorders)
- Articles, including preclinical research, reporting mechanisms of action, efficacy, or safety

Exclusion criteria
- Case reports or case series
- Studies with sample sizes of less than eight participants
- Non-peer reviewed literature or conference abstracts

Molecular and cellular foundations

Pharmacology

Classic psychedelics show diverse pharmacology, with binding profiles including various serotonergic, dopaminergic, and adrenergic targets.[7 8] However, their characteristic subjective effects are driven primarily by agonism at the 5-HT_{2A} receptor, which is densely expressed in cortical regions including the frontal, prefrontal, and visual cortices.[9-11] Human pharmacological studies show that 5-HT_{2A} receptor binding correlates with subjective effect intensity,[12] while pretreatment with the selective 5-HT_{2A} antagonist ketanserin almost entirely blocks these effects.[13-15]

Not all 5-HT_{2A} agonists are hallucinogenic. Endogenous serotonin and compounds including lisuride activate the receptor without altering consciousness.[16] This selectivity reflects biased agonism at 5-HT_{2A} receptors: hallucinogenic compounds including classic psychedelics preferentially activate the Gq signaling pathway, while non-hallucinogenic agonists preferentially activate the β-arrestin pathway, pointing to the vital role of biased agonism in driving subjective effects.[17] 5-HT_{2A} receptor agonism, whether it activates G_q or β-arrestin, also drives downstream changes in gene transcription, glutamate transmission, and AMPA (α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid) and NMDA (N-methyl-D-aspartate) receptor activation, processes that may underlie the evidenced pro-neuroplastic effects of serotonergic psychedelics.[18-20]

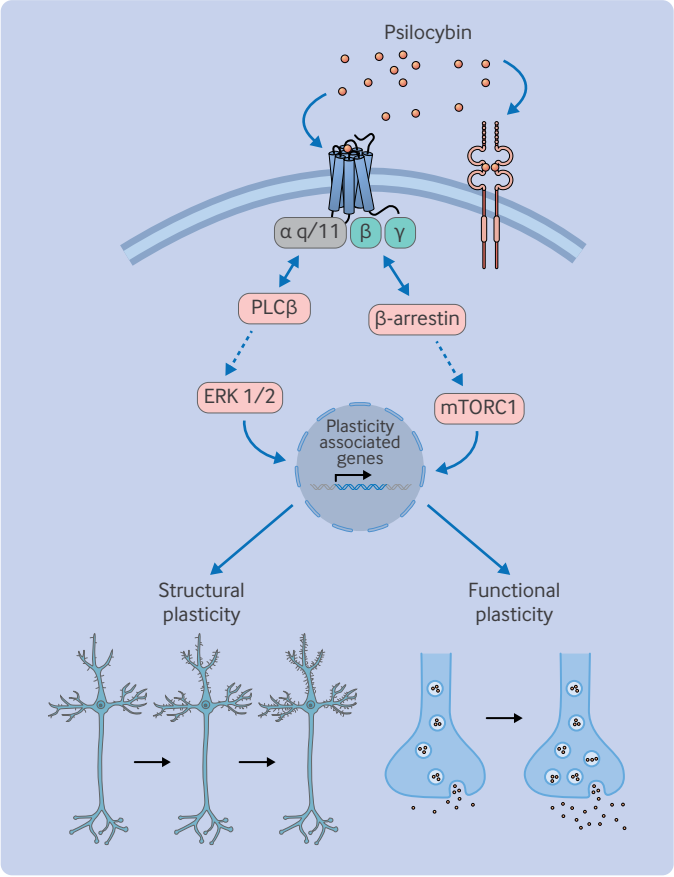

Fig 3 | Psilocybin binds at 5-HT_{2A} and TrkB (tropomyosin receptor kinase B) receptors on cell surface, starting various internal cascades that ultimately induce the transcription of plasticity associated genes that lead to structural and functional changes in plasticity. ERK=extracellular signal regulated kinase; mTOR=mammalian target of rapamycin; PLC=phospholipase C

Neuroplastic effects

Some preclinical work shows that psychedelics enhance neuroplasticity—the brain's capacity to modify its physical structure and functional connections in response to stimuli[21]—though data in humans remain sparse owing to methodological constraints. Given that dysfunctions in neuroplasticity have been implicated in various mental disorders,[22 23] these plasticity enhancing effects represent a candidate therapeutic mechanism. In animal models, psychedelics are understood to increase plasticity in various domains: structurally by creating new dendrites and spines; functionally by enhancing cell excitability and long term potentiation; and behaviorally by increasing cognitive and psychological flexibility—adaptive capacities are detailed further below—though whether these reflect the same underlying mechanisms remains to be established.

Molecular plasticity—At the molecular level, psychedelics have been shown to increase expression of plasticity associated genes, including those encoding for brain derived neurotrophic factor (BDNF) and components of the mammalian target of rapamycin (mTOR) pathway[24]—pathways also implicated in therapeutic responses to conventional antidepressants.[25] Initial work in single cell laboratory studies[26] showed this sequential relation: 5-HT_{2A} receptor activation leads to increased BDNF signaling through internal cascades and TrkB (tropomyosin receptor kinase B) receptors, the primary high affinity receptors for BDNF (fig 3). Using selective antagonists, the authors showed that blocking 5-HT_{2A} or TrkB prevented psychedelic induced increases in dendritogenesis, spinogenesis, and synaptogenesis. More recent evidence indicates psychedelics may directly modulate TrkB: both psilocin (the active metabolite of psilocybin) and

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LSD bind to TrkB with high affinity, with plasticity promoting effects persisting even when 5-HT_{2A} signaling is blocked.[27]

Direct evidence in humans remains methodologically challenging to obtain. While animal studies show increased central (brain) BDNF, peripheral BDNF levels in humans show no consistent changes after psychedelic administration,[28] though the association between peripheral and central BDNF remains incompletely understood. Establishing molecular plasticity changes in patients represents an important next step for the field.

Structural plasticity—These molecular changes drive structural modifications that could underlie long term therapeutic change. In mice, a single dose of psilocybin led to increases in dendritic spine size and density in the medial frontal cortex that persisted for at least one month,[29] mirroring the structural changes seen during successful learning and memory formation.[30] Similarly, in pigs,[31] psilocybin increased synaptic density marker SV2A in hippocampal and prefrontal cortex tissues, an effect which persisted for at least seven days.

Metaplasticity—Psychedelics may also induce metaplasticity—changes in the capacity for subsequent plasticity.[32] Unlike the direct structural changes described above, metaplasticity involves priming neurons to respond more readily to future plasticity inducing stimuli. Preclinical work shows serotonergic psychedelics create lasting alterations in neuronal responsiveness, increasing cortical excitatory transmission[33] and enhancing short term potentiation in hippocampal regions.[34] Thus, limited dosing may produce therapeutic benefits that persist well beyond acute drug effects—by creating windows of enhanced neural adaptability during which therapeutic insights, behavioral changes, and integration may work more readily to reshape neural circuits.

Circuit and network reorganization

Global network principles

Psychedelics markedly affect brain networks that underlie cognitive flexibility and emotional processing, suggesting therapeutic mechanisms for psychiatric disorders characterized by rigidly negative thought content[35] or disrupted emotional processing.[36] A finding replicated in several studies is globally decreased connectivity within brain networks alongside increased connectivity between networks.[37 38] Baseline functional connectivity patterns may predict the magnitude of psilocybin induced network changes[39] and antidepressant effect,[40] though larger studies are needed to validate potential biomarkers. Various networks and circuits crucial to causing psychiatric disease and its progression have been shown to change after psychedelic administration.

Default mode network: historical foundation

Early psychedelic neuroimaging focused on the default mode network (DMN)—a network of brain regions most active during rest and internally focused activities like self-reflection and daydreaming.[41] Aberrant DMN function is implicated across psychiatric conditions.[42] Functional magnetic resonance imaging shows psychedelics decrease within-DMN connectivity and increase its connectivity with other networks.[43] However, DMN changes lack specificity: alcohol, amphetamines, and selective serotonin reuptake inhibitors (SSRIs) also reduce within-network connectivity.[44-46] Therefore, investigation of more specific mechanistic frameworks is warranted.[4 8]

Cortico-striato-thalamo-cortical circuits

The cortico-striato-thalamo-cortical (CSTC) model (fig 4) has emerged as a key framework for understanding psychedelic effects. CSTC circuits regulate cortical input through thalamic gating of ascending sensory signals, with structural and functional abnormalities implicated across several psychiatric disorders.[47-49] The thalamic gating function is essential for maintaining perceptual coherence, attentional focus, and the suppression of irrelevant stimuli. LSD, through activation of 5-HT_{2A} receptors, particularly on cortical layer 5 pyramidal neurons, may relax these gating constraints, permitting greater sensory inflow to associative cortical areas,[50 51] and plausibly contributing to the perceptual alterations characteristic of psychedelic states. Psilocybin and LSD affect prepulse inhibition, an operational measure of CSTC dependent sensorimotor gating.[52 53] The interval dependent effect of psilocybin on prepulse inhibition—reducing prepulse inhibition at short delays but enhancing it at longer ones[54]—suggests selective modulation of thalamic gating rather than a global disintegration.

Emotional processing networks: clinical targets

Psychedelics also affect emotional processing circuits, which is particularly relevant given that many psychiatric disorders involve dysregulated emotional processing and altered threat perception. LSD has been shown to reduce amygdala reactivity to negative stimuli, with effects persisting beyond the acute drug state.[50] One week after taking psilocybin, healthy people showed reduced negative affect and decreased amygdala responses to facial affect stimuli.[55] In treatment resistant depression, improved affect recognition persists one month after taking psilocybin.[56] However, the temporal dynamics of these changes appear complex. Increased amygdala reactivity to fearful faces one day after psilocybin administration in people with depression correlated with positive clinical outcomes.[57] These findings suggest psychedelics may start a dynamic reorganization of emotional processing circuits that evolves over time, potentially normalizing negative biases and reducing emotional avoidance.[58 59]

REBUS model: computational perspective

The REBUS (relaxed beliefs under psychedelics) model provides a computational framework for

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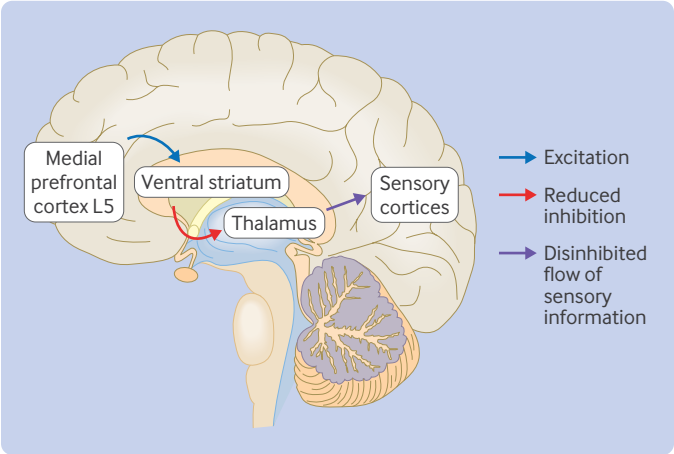

Fig 4 | Simplified representation of effects of classic psychedelics on cortico-striatal-thalamic functioning. 5-HT$_{2A}$ receptor activation in the medial prefrontal cortex layer 5 (L5) increases its excitatory output to the ventral striatum, which ultimately reduces inhibitory control over the thalamus. As a result, the normal gating function of the thalamus is disrupted and thalamocortical connectivity increased, enhancing sensory flow to the cortex and inducing perceptual alterations

understanding psychedelic therapeutic effects.[60] According to this model, psychedelics reduce the precision weighting of high level priors, diminishing the influence of entrenched, top-down beliefs on perception and cognition while allowing bottom-up sensory information greater influence. Supporting this framework, preliminary neurophysiological evidence in healthy volunteers (n=20) showed muted neural markers of surprise to unexpected stimuli after 75 µg of LSD, suggesting a loosening of previous expectations.[61]

This mechanism may explain the apparent transdiagnostic efficacy of psychedelics. Many psychiatric disorders involve maladaptive internal models. Therapeutic loosening of these pathological predictions may create windows for adaptive belief updating during the integration phase of treatment.

REBUS also explains the sensitivity of psychedelics: with high level predictions weakened, environmental inputs gain greater influence on experience and belief formation. This suggests that the post-acute integration phase, when people work with therapists to process insights and consolidate new perspectives, may be particularly effective because of enhanced environmental receptivity.

While theoretically compelling, bridging implementational findings (such as altered DMN connectivity) to computational mechanisms (such as modified predictive processing) remains challenging, and direct empirical validation of REBUS predictions in clinical populations has yet to be shown.

Methodological considerations

The association between brain network changes and clinical outcomes remains difficult to interpret, in part because the effects of psychedelics on brain activity appear to be highly context dependent. Task engagement attenuates psychedelic induced brain network changes,[62] highlighting a broader methodological challenge: resting state findings may not capture how drugs modulate brain activity during clinically relevant cognitive and emotional processes. Furthermore, considerable variability in analytic approaches across resting state studies complicates cross-study comparisons. Future work using standardized task based paradigms and harmonized imaging methods will be crucial for understanding how circuit changes relate to therapeutic outcomes.

These circuit level reorganizations may provide the neurobiological substrate for the cognitive and psychological changes observed during therapeutic response, though the precise relations between network changes and clinical outcomes remain to be clarified.

Cognitive and behavioral outcomes

Cognitive flexibility

Human evidence for psychedelic enhanced plasticity stems from behavioral measures, particularly post-acute increases in cognitive flexibility—the ability to adaptively switch between tasks, mental frameworks, and strategies as the environment demands. A single dose of psilocybin (25 mg) enhanced cognitive flexibility in patients with major depressive disorder (n=24).[63] In this open label trial, psilocybin improved set shifting performance by reducing perseverative errors, the tendency to persist in following outdated response rules, for at least seven days. Importantly, these effects were independent of mood change, suggesting a distinct neuroplastic mechanism. Complementary observational data from a prospective survey of 657 people using psilocybin mushrooms showed self-reported cognitive flexibility increased significantly at two to four weeks (standardized mean difference 0.23) and two to three months (0.22) after use, with greater "mystical experiences" predicting larger improvements.[64] Although promising, these findings require replication in larger clinical samples.

Emotional processing and behavioral engagement

Beyond the circuit level changes described above, psychedelics produce measurable behavioral improvements in emotional processing. One study of rats given psilocybin showed increased engagement in reward seeking tasks through an optimism bias,[65] an effect that persisted for 14 days and parallels behavioral activation interventions in depression treatment. In an open label trial examining psilocybin for treatment resistant depression, participants (n=17) showed baseline impairments in facial emotion recognition speed compared with controls. One month after two psilocybin sessions (10 mg and 25 mg), this deficit was largely remediated, with faster emotion recognition correlating with greater reductions in anhedonia,[56] but not with overall depression scores. These findings suggest potential restoration of reward related behavioral responses disrupted during depression.

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Subjective and psychological processes

The relation between various subjective aspects of psychedelic experiences and therapeutic outcomes remains contentious.[66 67] Studies have reported associations between mystical-type experiences and transdiagnostic clinical improvement in cancer related distress,[68 69] treatment resistant depression,[70] and substance dependencies.[71 72] The mystical-type experience, as measured by the mystical experience questionnaire, encompasses feelings of unity, positive mood, transcendence of time and space, and ineffability. Similar constructs include the "oceanic boundlessness" factor of the five dimensional altered states of consciousness scale (assessing feelings of connection, positive mood, and spiritual experience), and the ego dissolution inventory (measuring self-world boundary dissolution).[73] However, causal relevance remains unclear.[74] Crucially, both therapeutic benefit and subjective effects are thought to depend on 5-HT_{2A} agonism, complicating experimental dissociation.

Subjective measures of acute effects encounter phenomenological validity challenges.[73] Acute cognitive deficits may impair recall accuracy, while differences in interpreting terms like "spiritual" or "unity" may introduce systematic measurement error. Pronounced subjective effects may also induce "halo effects": participants who have had an intensely positive or meaningful experience may rate all positive descriptors highly, rather than discriminating between specific experiential qualities.

Certain other psychological processes show consistent associations with outcomes. Emotional breakthrough—the confrontation of typically avoided emotional material[75]—was the strongest subjective predictor of antidepressant response at three week follow-up in the largest psilocybin trial (n=233; r=−0.512).[76] A systematic review found that psychological insight—new understanding about oneself, relationships, or existential concerns—was significantly higher in psychedelic than placebo conditions in 93% of controlled studies making this comparison, with stronger outcome associations than mystical experiences in 55% of comparative studies.[77]

These processes may work sequentially: emotional breakthrough during acute effects facilitates psychological insight post-acutely, enabling revised self-understanding and improved adaptive responding to future stressors. Increases in psychological flexibility—the ability to adapt behavioral strategies based on changing demands rather than immediate emotional reactions or rigid behavioral patterns[78]—mediate psychedelic induced improvements across depression, anxiety, and trauma symptoms.[78 79]

These findings show how psychedelic induced neuroplasticity and circuit reorganization may translate into clinically meaningful change. Enhanced cognitive flexibility and normalized emotional processing represent measurable improvements in core deficits across psychiatric disorders, while increased behavioral activation directly addresses motivational deficits characteristic of depression. More research is needed on aspects of the acute subjective effects and persisting effects of psychedelics.

MDMA mechanisms of action in PTSD

MDMA has distinct pharmacology and therapeutic applications compared with those covered above relating to classic psychedelics. A synthetic drug known for its stimulant and empathogenic effects, enhancing energy, emotional connection, and mood, MDMA is being tested primarily for PTSD using a more actively engaged psychotherapy model than the non-directive support typical of trials with classic psychedelics (though some classic psychedelic studies also include psychotherapy; fig 1).

Pharmacological mechanisms

Like classic psychedelics, MDMA binds to several serotonin receptors. Ketanserin pretreatment attenuates the characteristic emotional and perceptual effects of MDMA, suggesting that 5-HT_{2A} receptor activation contributes to these effects, as with classic psychedelics.[80] However, MDMA acts primarily by triggering the release of monoamines through their respective transporters, predominantly serotonin, followed by norepinephrine and dopamine. Serotonin transporter inhibition is particularly linked to positive mood effects, while norepinephrine transporter mediated effects contribute to stimulant-like properties.[81]

Neurophysiological mechanisms

These acute pharmacological actions trigger changes in emotion processing neural circuits. Initial neuroimaging in healthy adults shows reduced amygdala blood flow and increased amygdala-hippocampal connectivity after MDMA.[82] This increased coupling, observed at rest, may reflect enhanced integration of emotional and contextual memory processes, though the direction of information flow cannot be inferred from resting state data. This circuit is crucial for emotional memory formation, contextualizing affective information, and fear processing.[83 84] Dysfunction in amygdala-hippocampal connectivity is implicated in causing PTSD and anxiety disorders.[85] Patients with PTSD typically show reduced connectivity compared with controls.[86] Normalization of amygdala-hippocampal communication could theoretically enhance adaptive processing of emotional stimuli and emotional memory consolidation in PTSD. However, the therapeutic relevance remains unclear. The single available neuroimaging study of patients with PTSD following MDMA assisted therapy (n=9) found increases in amygdala-hippocampal functional connectivity that approached but did not reach statistical significance. Given the small sample, underpowered even by functional magnetic resonance imaging standards, this result

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permits little confident inference.[87] Like classic psychedelics, MDMA appears to decrease within-network functional connectivity of the default mode network,[88] although it has a less pronounced effect on between-network functional connectivity.[89]

Recent work suggests that MDMA reopens a critical period for social reward learning through oxytocin receptor activation in the nucleus accumbens. This plasticity requires both serotonin transporter activation and a social context, suggesting a mechanistic basis for MDMA's enhancement of therapeutic relationships.[90] One study found that LSD and psilocybin also show context dependent social learning effects,[91] suggesting another potentially shared mechanism that may contribute to therapeutic effects across MDMA and classic psychedelics. However, a coordinated five laboratory study was unable to replicate psilocybin induced reopening of this critical period, though protocol differences limit direct comparison with the original findings.[92]

Psychological and therapeutic mechanisms

In line with early clinical observations,[93] MDMA produces acute effects that likely facilitate psychotherapy: selective impairment of negative emotion recognition,[94-96] increases in trust, cooperation, and sociability,[97 98] and decreased concern about negative evaluation by others.[99] Prosocial changes partly reflect release of oxytocin, a neuropeptide implicated in parent-infant bonding, social affinity, and trust,[100] with these effects blunted in patients with oxytocin deficiency secondary to hypothalamic dysfunction.[101] Oxytocin mediated prosociality may enhance therapeutic alliance, while acutely reduced fear may enable trauma processing without overwhelming arousal.[102]

The reprocessing of aversive stimuli is central to evidence based PTSD treatments.[103 104] Preclinical models show that MDMA enhances fear extinction and disrupts fear memory reconsolidation.[105 106] In mice, fear extinction enhancement requires both serotonin transporter and 5-HT_{2A} receptor activation,[107] with MDMA treatment increasing BDNF expression in the amygdala,[105] suggesting that synaptic plasticity may mediate therapeutic effects. Although patients with PTSD show diminished fear extinction,[108] human studies show that MDMA (125 mg) enhances fear extinction and its retention as measured by skin conductance response, but not fear potentiated startle.[109]

How these mechanisms interact remains unclear. Both MDMA and classic psychedelics appear to enhance neuroplasticity and alter emotional processing through serotonergic mechanisms, but through distinct pathways. Striking alterations in consciousness using classic psychedelics may facilitate emotional breakthrough and subacute cognitive flexibility, while MDMA's prosocial and anxiolytic properties specifically enhance processing of traumatic memories by supporting therapeutic alliance. This mechanistic divergence is reflected in distinct psychotherapeutic approaches studied to date. Classic psychedelics typically use non-directive support during sustained periods of internally directed experience, often involving vivid imagery, autobiographical memory, and altered self-perception. By comparison, MDMA assisted therapy (principally the Lykos protocol for PTSD, though other exposure protocols are being developed to pair with MDMA from more evidence based psychotherapies) involves active therapeutic dialogue and verbal processing through the drug session.

Clinical evidence

Although an integrative understanding of therapeutic mechanisms remains to be elaborated, evidence suggests that classic psychedelics and MDMA may be effective in treating diverse neuropsychiatric conditions. Evidence quality remains variable, but maturing evidence bases are emerging for specific indications.

This review now examines the clinical evidence, focusing on the most extensively studied applications: MDMA assisted therapy for PTSD, and psilocybin and other classic psychedelics in depressive disorders. We then survey evidence on substance use disorders and psychiatric symptoms secondary to life threatening illness. Our scope prioritizes the most rigorous trials, emphasizing larger samples when possible.

Post-traumatic stress disorder

MDMA's therapeutic potential in PTSD has been investigated through a systematic research program spanning several clinical trials. To date, approximately 289 participants have been enrolled in randomized, placebo controlled trials of MDMA assisted therapy for PTSD. While this sample size provided adequate statistical power given the large effect sizes observed in exploratory studies, it is smaller than the thousands of participants usually enrolled before drug licensing. Moreover, these trials have typically screened out around 90% of potential participants, using exclusion criteria more restrictive than typical PTSD research—particularly any personality disorders, or non-alcohol or cannabis substance use disorder within 12 months—compared with 13% of PTSD psychotherapy trials that exclude personality disorders, and 56% of trials that impose no substance use restriction.[110] Given the high rates of substance use comorbidity in patients with PTSD,[111 112] these restrictions raise questions about generalizability to broader clinical populations.

The first phase 3 trial (MAPP1) compared MDMA assisted therapy (80-120 mg) with placebo with therapy in 90 participants with severe PTSD.[113] At 18 weeks, MDMA produced significantly greater reductions in CAPS-5 (clinician administered PTSD scale for DSM-5) scores than placebo (difference 11.9 points, 95% confidence interval (CI) 6.3 to 17.4, Cohen's d=0.91, P<0.0001). Although 67% of MDMA recipients no longer met PTSD criteria versus 32% of those receiving placebo, high rates of functional unblinding (when psychedelics' obvious

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effects undermine blinding integrity) complicate interpretation—32% of participants reported previous MDMA experience, and higher dropout in the placebo group suggests possible expectancy effects.

A second phase 3 trial (MAPP2) extended these findings in a more diverse population, randomizing 104 participants to MDMA (n=53) or placebo (n=51) with identical therapy.[114] The sample included 33.7% non-white and 26.9% Hispanic or Latino participants. The results largely replicated MAPP1. At six to eight weeks after the trial's third and final drug assisted session, MDMA showed greater reductions in CAPS-5 scores (difference −8.9 points, 95% CI −13.7 to −4.1, Cohen's d=0.7, P<0.001) and higher rates of PTSD remission (46.2% *v* 21.4%). Treatment effects were consistent regardless of symptom severity, trauma type, or presence of the dissociative subtype of PTSD. MDMA assisted therapy also showed superiority on functional impairment measured by the Sheehan disability scale (difference −1.20, 95% CI −2.26 to −0.14, Cohen's d=0.4, P=0.03). However, functional unblinding remained problematic: 94% of MDMA recipients correctly guessed their treatment assignment. While differential dropout rates persisted, they were less pronounced (1.9% MDMA *v* 15.7% placebo).

A recent meta-analysis confirms the efficacy of MDMA assisted therapy versus placebo across trials (Hedges' g=−1.53, 95% CI −2.56 to −0.49 for CAPS-5 reduction), with significantly higher rates of loss of diagnosis and remission.[115] Interpretation of these findings is complicated by several methodological factors.

Functional unblinding represents a primary concern. Although blinding integrity is rarely assessed in clinical trials generally,[116 117] the extent of unblinding in MDMA assisted therapy trials is notable. Importantly, recorded effect sizes in these trials exceed typical placebo response rates in PTSD trials,[118 119] suggesting therapeutic effects beyond expectancy. However, recent quantitative analysis[120] estimating unmasking bias in ketamine trials (standardized mean difference 1.1) noted that this exceeds MDMA assisted therapy effect sizes (standardized mean difference 0.7-0.9), suggestsing that functional unblinding may account for observed effects, though important differences between these interventions preclude direct comparison. Supporting this concern, significantly different dropout rates between active and placebo groups in MAPP1 may reflect nocebo effects in the placebo group rather than inflated expectations in the treatment group.

Additional factors complicate interpretation. High rates of previous MDMA experience (32% and 46% in the phase 3 trials) have raised concerns about expectancy effects, particularly given that recruitment has been reported to draw "heavily from the existing community of those interested and involved in the use of psychedelics for possible psychological benefits."[121] However, empirical analysis presented to the FDA advisory committee showed no significant association between previous MDMA use and treatment outcomes.[122 123] More critically, blinded, placebo controlled data beyond two months of treatment are lacking, limiting our understanding of long term efficacy and safety. These methodological concerns, alongside other contested aspects of the evidence base likely contributed to the FDA's decision not to license MDMA assisted therapy in August 2024.

Depressive disorders

Depression is the most studied disorder in psychedelic research. Initial controlled evidence for psilocybin's antidepressant effects came from a randomized waitlist controlled trial of 24 participants with moderate to severe major depressive disorder.[124] Participants received two sessions of psilocybin therapy (20 mg/70 kg bodyweight, then 30 mg/70 kg bodyweight) and depression severity was assessed using the GRID Hamilton depression rating scale (GRID-HAMD) by clinicians who were masked to treatment arm. At five weeks, the treatment produced large reductions in depression severity compared with the waitlist (Cohen's d=2.6, 95% CI 1.5 to 3.7, P<0.001). At four weeks after both sessions, 71% had a clinically significant response to the intervention (≥50% reduction in GRID-HAMD score), and 54% were in remission (≤7 GRID-HAMD score). A 12 month follow-up study of this sample found treatment response and remission to be 75% and 58%, respectively,[125] although a third of participants had started using antidepressants during that time.

In a larger randomized controlled trial (n=104) using a niacin active placebo, single dose psilocybin (25 mg) produced greater mean reductions in Montgomery-Åsberg depression rating scale (MADRS) scores after six weeks (between group difference −12.3, 95% CI −17.5 to −7.2).[126] By this time point, more patients in the psilocybin group (42%) than the niacin group (11%) had a sustained antidepressant response (≥50% MADRS reduction).

The first head-to-head comparison against an established antidepressant[127] randomized 59 participants with depression to receive psilocybin or escitalopram, each with psychological support. The escitalopram group received 1 mg psilocybin (active placebo) on days 1 and 21 plus daily escitalopram (10 mg increasing to 20 mg over six weeks), while the psilocybin group received 25 mg doses on days 1 and 21 plus daily placebo capsules (likewise doubling over six weeks). While no significant difference was found on the primary outcome measure (QIDS-SR-16—quick inventory of depressive symptomatology-self-report-16 items) at six weeks, psilocybin showed advantages on several secondary measures of depression and wellbeing—though these were not corrected for multiple comparisons. A six month follow-up found significant differences in favor of psilocybin on measures of work and social functioning (work and social adjustment scale: unstandardized coefficient estimate −7.46, 95% CI −12.4 to −2.47, P<0.001), as well as meaning in

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life (meaning in life questionnaire: 4.86, 0.67 to 9.05; P=0.021).[128]

One large trial of psilocybin (n=233) compared single doses of psilocybin (25 mg or 10 mg) with an active placebo-like control condition of very low dose psilocybin (1 mg) in treatment resistant depression.[129] Each treatment group also received limited psychological support. This was a positive trial—high dose psilocybin (25 mg) reduced depression more than the 1 mg psilocybin control condition (−6.6 MADRS difference between groups, 95% CI −10.2 to −2.9), with an odds ratio of response at three weeks of 2.9 (95% CI 1.2 to 6.6) for 25 mg and 1.2 (0.5 to 3.0) for 10 mg. Although, as in the study by Carhart-Harris and colleagues,[127] high dose psilocybin was associated with greater improvements in a range of secondary measures, the favorable odds ratio of antidepressant response for the 25 mg dose was not sustained at 12 weeks after administration.[130]

More recent work has begun bridging the gap to clinical implementation, albeit in smaller and open label studies. An open label study (n=29) included previously excluded populations (bipolar II depression (n=4); comorbid personality disorder (n=9)) and investigated repeated dosing.[131] Initial psilocybin dosing (25 mg) produced significantly greater MADRS reductions than waitlist controls at two weeks (Hedge's g=1.1, 95% CI 0.3 to 1.8, P=0.005). Upon relapse, 17 participants received a second dose, and five received a third, between 10 and 20 weeks after treatment. Successive doses were associated with cumulative symptom improvement. No treatment emergent mania or psychosis occurred in participants with complex presentations.

Although these findings indicate psilocybin's antidepressant potential, key limitations include the exclusion of common comorbidities, outcomes assessed only to 12 weeks, and inadequate assessment or maintenance of blinding. The optimal dosing frequency and appropriate psychotherapeutic support remain to be determined.

Substance use disorders

Classic psychedelics also show promise for substance use disorders.[132] After publication of several research trials in substance use disorders in the first wave of psychedelic research,[133 134] we review clinical evidence from modern era trials, all of which have embedded drug administration within comprehensive psychosocial treatment programs.

For alcohol use disorder, a multisite randomized controlled trial (n=95) compared two doses of psilocybin (25 mg/70 kg bodyweight and 25-40 mg/70 kg bodyweight) with 50 mg and 100 mg of the sedative antihistamine diphenhydramine.[135] Drug treatment was supported by 12 psychotherapy sessions, including aspects of motivational enhancement therapy, cognitive behavioral therapy, and bespoke support sessions before and after each drug session. While both groups reduced their drinking, psilocybin showed superiority at 33-36 weeks for biomarker confirmed abstinence (odds ratio 2.84, 95% CI 1.17 to 6.89, P=0.02) and absence of heavy drinking days (2.5, 1.08 to 5.76, P=0.03). The psilocybin treatment group reported less than half of the heavy drinking days of the control group. However, there were high levels of functional unblinding (about 90%) in this study.

More recent trials have tested psilocybin in different contexts of alcohol use disorder with mixed results. A randomized controlled trial in patients with severe alcohol use disorder and comorbid depression (n=30) found that two doses of 25 mg psilocybin (*v* 1 mg control) significantly improved abstinence rates at 12 weeks (55% *v* 11%, P=0.03).[136] However, a separate randomized controlled trial (n=37) investigating single dose psilocybin (25 mg) for relapse prevention after withdrawal found no significant differences in abstinence duration or alcohol use compared with a mannitol placebo.[137]

Tobacco use disorder may also be amenable to psychedelic based treatments. An open label pilot study (n=15) investigated two to three doses of psilocybin (20 mg/70 kg bodyweight; 30 mg/70 kg bodyweight) within a 15 week psychosocial intervention including cognitive behavioral therapy for smoking cessation.[138] Abstinence rates were striking: 80% at six months and 60% at long term follow-up (mean 30 months), notable in a cohort averaging 19 cigarettes a day for 31 years, and six previous attempts to quit.[138 139]

Psychological distress associated with life threatening illness

Psychological distress in life threatening illness was an early focus of modern clinical psychedelic research, building on findings from the 1950s-70s.[140 141] After pilot studies using LSD[142] and psilocybin[143] demonstrated feasibility, two pivotal randomized crossover trials provided controlled evidence for efficacy.

Two trials evaluated psilocybin treatment in people who had received a life threatening diagnosis. In one study,[69] 51 patients with cancer and depression or anxiety symptoms, or both (including adjustment disorders) received high dose (22 or 30 mg/70 kg bodyweight) and low dose (1 or 3 mg/70 kg bodyweight) psilocybin. Five weeks after the first session, participants receiving high dose psilocybin showed significantly greater improvements in depression, anxiety, quality of life, and optimism. By six month follow-up, clinician rated measures showed persistent reductions in depression (GRID-HAMD-17) and anxiety (Hamilton anxiety rating scale) across all participants (Cohen's d=2.98 and 3.40, respectively). However, limited blinding assessment and the highly selected, demographically homogeneous sample (94% white, 53% postgraduate educated) restrict the generalizability of findings. Concurrent work[68] compared psilocybin (0.3 mg/kg bodyweight) with niacin (250 mg) in 31 patients with cancer related anxiety and depression (predominantly adjustment disorders). Seven

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weeks after the first dose (ie, before crossover), the psilocybin first group showed significantly reduced anxiety and depression across several validated self-report measures (Cohen's d=0.82-1.36). Although 55% reported previous psychedelic use, this did not significantly interact with outcome measures. Study staff correctly guessed group assignment for 97% of participants, whose blinding was not assessed. After six months, participants showed reduced depression, anxiety, demoralization, and hopelessness compared with baseline. Long term follow-up of surviving participants (n=15) found more than 60% reported enduring antidepressant or anxiolytic effects at 4.5 years, independently of cancer remission status.[144] Only 8% received further psychotherapy or pharmacotherapy for cancer related distress after their trial participation.

Safety and monitoring

While the trials discussed above provide promising evidence of efficacy, establishing the safety profile of psychedelic therapies is crucial for clinical translation. Below, we examine adverse events associated with psychedelic administration and safety practices in clinical trials.

Adverse events

Clinical trials of psilocybin and MDMA show manageable safety profiles under controlled conditions, with most adverse effects being transient and self-limiting. Although reports of serious or non-serious adverse events requiring intervention are rare, they do occur. This underscores the importance of careful monitoring and reporting, a priority for the field given current methodological variability in adverse event assessment and reporting.[5 145]

The physiological effects of psilocybin, LSD, and MDMA are well characterized.[7] While the clinical environment can limit risk factors associated with recreational psychedelic use,[146 147] meta-analyses of available adverse event data in psilocybin[148] and MDMA[145] trials report high levels of adverse events that are largely tolerable, resolve within 48 hours, and do not lead to study withdrawal.

Some minor physiological adverse events are relatively well characterized and expected. For psilocybin versus comparators (including low dose psilocybin, niacin, and escitalopram; six trials, n=528), acute effects include increased blood pressure (relative risk 2.29), headache (1.99), nausea (8.85), dizziness (5.81), and anxiety (2.27, affecting 4-26% of participants), with lower rates of thought disorder (about 5%) and paranoia (about 2%).[148] In placebo controlled MDMA assisted therapy trials for PTSD (three trials, n=214), prominent effects include hyperhidrosis (relative risk 6.37), chills (7.36), decreased appetite (4.12), mydriasis (13.41), nystagmus (6.10), and blurred vision (6.52).[115]

Psychedelic dosing sessions can involve powerful emotional experiences, including intense fear, overwhelming grief, or severe anxiety episodes that, while typically not meeting criteria for serious adverse events, can be highly distressing and require skilled therapeutic management. Psychological adverse events are relatively common. Most psychological reactions are mild and respond to psychological support, with pharmacological intervention required in approximately 0.5% of cases.[149] Subacute effects include headache in 7-67% of psilocybin recipients,[150] while MDMA recipients commonly report fatigue, low mood, and irritability for three to seven days after administration (about 40% in phase 3 trials[113 114]). However, enduring psychological and psychiatric effects, while rare, can be serious and warrant attention. Among 584 outpatient participants receiving high dose psilocybin, 23 experienced delayed serious adverse events including suicidal ideation and behavior.[5] Although prospective participants judged to be at clinically significant risk of suicide were excluded from one large psilocybin trial (n=233), the trial recorded three participants in the 25 mg group reporting suicidal behavior, two in the 10 mg group reporting suicidal ideation or self-injury, and one in the 1 mg group engaged in self-injury.[129] One completed suicide occurred in the control group (participant receiving 1 mg psilocybin) of a cancer related distress trial.[69] Although these events occurred in clinically complex populations and may be attributable to other factors, careful investigation of potential mechanistic relations, and development of specific risk monitoring and management protocols, are warranted.

The association between psychedelic use and psychotic or manic symptoms represents a frequently cited safety concern. A meta-analysis of nine studies (totaling 7931 sessions with psilocybin or LSD)[151] found low incidence rates of psychedelic induced psychosis: 0.4% in general populations and 0.2% in healthy people, though evidence quality was generally low with substantial heterogeneity among studies. Among those experiencing psychedelic induced psychosis, 13.1% later developed schizophrenia.

Bipolar I disorder emerges as a critical risk factor, with several studies showing increased manic or psychotic symptoms after psychedelic use.[152 153] Some evidence paradoxically suggests protective effects for psychotic disorder histories, but replication is needed.[153 154] Context and frequency of use appear relevant. Supervised clinical settings dramatically reduce risk compared with illegal use, which correlates with increased symptoms.[151 152]

These findings support current clinical exclusion criteria for bipolar I disorder and schizophrenia, while highlighting the importance of careful screening, professional supervision, and controlled administration contexts for therapeutic applications.

Drug-drug interactions

Current protocols require pretreatment drug review. Coadministration of dopamine D_2 antagonists such as haloperidol with classic psychedelics may be inadvisable. In a controlled trial, haloperidol did not attenuate psilocybin's effects but exacerbated

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Box 2: Areas of psychedelic assisted therapy needing evidence based guidance

- Minimum qualifications and training requirements for therapists
- Essential components and duration of psychological support with adherence and fidelity ratings
- Facility specifications, standardized documentation, and monitoring protocols
- Management of common clinical scenarios (eg, acute psychological distress)
- Patient selection criteria balancing safety with accessibility
- Modifications needed for patients with common psychiatric comorbidities
- Standards for outpatient follow-up and adverse event monitoring

anxiety and thought disturbance, suggesting potential for worsened psychological responses.[15] By contrast, 5-HT_{2A} antagonists such as risperidone must be discontinued because they block the primary mechanism of action. SSRI continuation may be acceptable—a controlled trial (n=27) found escitalopram pretreatment neither compromised psilocybin's effects nor increased adverse effects.[155] An open label trial of 19 participants with treatment resistant depression who continued SSRI treatment showed antidepressant effects comparable to those seen in trials requiring drug discontinuation.[156] Caution is warranted with lithium because retrospective and anecdotal reports from recreational use suggest an increased risk of seizures when combined with classic psychedelics.[157]

For MDMA, the impact of SSRI discontinuation on MDMA assisted therapy presents distinct considerations given shared action at the serotonin transporter, with pretreatment blunting MDMA's subjective effects.[158] Interactions with antipsychotics also warrant consideration. The D_2 antagonist haloperidol has been found to shift MDMA's characteristic euphoric profile to a dysphoric one, with reduced wellbeing and increased anxiety reported.[159]

Importantly, systematic reviews of adverse event reporting in trials of classic psychedelics[5] (214 trials, n=3504) and MDMA assisted therapy[145] (eight trials, n=298) found highly variable and inadequate reporting across the literature. The latter study found that no randomized controlled trial of MDMA assisted therapy had adequate adherence (>70%) to the CONSORT Harms 2022 guidelines for reporting harms in clinical trials.[160] Moreover, 56% of non-serious and 31% of serious adverse events recorded on ClinicalTrial.gov registers were not reported in trial related publications. A systematic review and meta-analysis of experimental and clinical studies in early and recent psychedelic research found that 53.3% of included studies reported at least some adverse event data. Of these, only 24% used a systematic approach to adverse event assessment.[5]

Inadequate adverse event reporting is not unique to psychedelic research, but widespread across psychotherapeutic, psychiatric, and medical trials.[161-163] Although psychedelic trials exceed mean reporting standards in medical treatment contexts,[5] the prospect of systematic underdetection of adverse events, with associated medical and ethical risks, should be addressed urgently.[164] Future priorities include standardizing drug continuation protocols and implementing comprehensive adverse event assessment to ensure accurate risk evaluation and evidence based safety guidelines (box 2).

Ongoing considerations

Problem of functional unblinding

Functional unblinding has prompted a gear shift in trial design. The FDA's 2023 draft guidance for psychedelic research explicitly acknowledges this challenge, recommending functional unblinding assessments and independent outcome assessors.[165] Recent phase 3 psilocybin trials for treatment resistant depression—and those under way—use centralized raters masked to treatment assignment and assessment time points, attempting to reduce rater bias even when participants correctly guess their treatment. Trial sponsors reported that the first such trial met its primary endpoint (25 mg *v* placebo: −3.6 MADRS at week 5, P<0.01, n=258)[166] with smaller effects than phase 2 trials.

The complex association between expectancy and outcomes further complicates interpretation. One trial comparing psilocybin with escitalopram found no correlation between psilocybin expectancy and outcomes,[167] while an LSD crossover study showed better results when the active drug was given first, despite identical acute effects.[168] Participants who received placebo before LSD showed diminished therapeutic response to LSD, highlighting complex interactions between drug timing, psychological context, and therapeutic outcomes that current models do not fully capture. These findings suggest that while perfect blinding may be unattainable, systematic measurement of expectancy and unblinding patterns could help distinguish pharmacological from psychological contributions.

Improving scalability

Standard protocols requiring 15-25 hours of therapeutic contact pose significant implementation barriers. The field is exploring two primary responses, each with distinct tradeoffs.

Shorter acting compounds—These drugs promise reduced clinical burden. Trial sponsors reported that a recent multisite trial of intranasal 5-MeO-DMT (5-methoxy-N,N-dimethyltryptamine, n=193) found sustained depression improvements through to week 8 with 90 minute discharge times.[169] This finding fits existing two hour clinic paradigms established by treatments like Spravato. Whether ultrashort protocols maintain efficacy while ensuring safety requires systematic investigation.

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Group delivery models—These models show preliminary feasibility. At a community cancer center, an open label study (n=30) combined individual and group preparation or integration with a single 25 mg psilocybin dose for patients with cancer and major depression, achieving 80% sustained response and 50% remission at 8 weeks.[170] The HOPE trial (n=12) advanced this approach by conducting group psilocybin (25 mg) sessions using partitioned spaces, reducing therapist-patient ratios to 1:1.[171] Although these designs may improve scalability, it is unclear whether efficiency gains compromise therapeutic alliance, a factor increasingly understood to be a mediator of treatment efficacy.[172 173]

Expanding indications

Psychedelic trials are expanding into non-psychiatric indications, including fibromyalgia, cluster headache, and Parkinson's disease. Recent pilot and early phase randomized controlled trials (typically n<30) report promising signals in pain modulation, motor symptom improvement, and quality of life outcomes.[174-176]

Critical gaps requiring resolution

Despite methodological advances, several issues persist.

Durability remains unknown—Controlled data beyond 12 weeks are lacking for most indications. Without understanding relapse rates, optimal retreatment intervals, or long term safety, implementation planning lacks solid evidentiary foundation.

Mechanism uncertainty impedes optimization—Trials are yet to establish the process by which potential contributions to outcomes such as neuroplasticity, altered neural networks, or psychological insights occur. Enduring mechanistic ambiguity prevents rational protocol refinement.

Health equity considerations are underdeveloped—Current trial populations are predominantly white, educated, and high income,[177-179] raising questions about generalizability across diverse populations.[180] Future implementation must address cultural adaptation of therapeutic protocols, training diverse therapist workforces, and ensuring accessibility in underserved communities where mental health inequalities are most pronounced.

Economic sustainability lacks clear models—Psychedelic therapies involve high upfront costs for therapist time, infrastructure, and clinical monitoring, making them more like surgery and gene therapies than standard pharmacotherapy. Innovative payment approaches for gene therapies including outcome based agreements could, if adapted appropriately, mitigate payer risk while improving patient access. Early health economic analyses (eg, for MDMA assisted therapies for PTSD)[181 182] suggest favorable long term value, but assume sustained benefits not yet shown beyond limited follow-up periods.

Emerging treatments

Beyond the indications discussed above, several phase 3 trials are now under way. Two trials of MM120, a formulation of LSD, are recruiting for generalized anxiety disorder (ClinicalTrials.gov NCT06741228, NCT06809595) following positive phase 2b results that led to FDA breakthrough therapy designation. A phase 3 trial of MM120 for major depressive disorder (NCT06941844) has also begun dosing. Additionally, trials are expanding into populations with psychiatric comorbidities typically excluded from psychedelic research. Phase 2 studies are examining psilocybin for comorbid major depression and alcohol use disorder (NCT04620759) and MDMA for comorbid PTSD and alcohol use disorder (NCT05709353)—populations that better reflect clinical reality. For tobacco use disorder, two randomized controlled trials are comparing psilocybin with nicotine replacement therapy, and with niacin placebo (NCT01943994, NCT05452772).

Guidelines

Current clinical guidance primarily derives from research protocols[146 183] and FDA recommendations,[165] though comprehensive guidelines await regulatory approval and real world implementation data. Core principles emphasize licensed healthcare providers as session monitors, an on-call physician available within 15 minutes, comprehensive protocols for medical and psychiatric screening and the management of adverse events, and provision of a supportive physical environment. Research into patient experiences of treatment protocols remains limited, though emerging work suggests implementation challenges around managing discomfort during dosing and navigating treatment transitions.[184]

Several critical areas lack evidence based guidance (box 2), and the development of comprehensive clinical guidelines will require systematic assessment of adverse events and their management, comparative effectiveness studies of different support models, and implementation research in diverse healthcare settings.

Conclusion

This review reveals a rapidly evolving field with great promise and significant challenges. Despite substantial media coverage, it is imperative that perspectives of the risk-benefit profile of psychedelics remain evidence based.[1] Although early phase trials of psychedelic assisted therapies have generated considerable enthusiasm, with promising signals across several psychiatric indications, the recent FDA advisory committee decision on MDMA highlights the challenges in translating these interventions into approved treatments. The evidence base, though expanding rapidly, is characterized by methodological limitations complicating interpretation: functional unblinding, variable psychotherapeutic support, and homogeneous trial populations.

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Despite these limitations, the strongest evidence supports psilocybin for depression and MDMA for PTSD, with emerging data for substance use disorders and psychological distress secondary to life threatening illness. Across indications, therapeutic effects appear to occur through complementary biological and psychological mechanisms: enhanced neural plasticity, reorganization of emotional processing networks, and psychological flexibility may work in concert to enable sustained therapeutic change. While generally well tolerated under controlled conditions, safety assessment requires more systematic approaches, particularly relating to delayed psychological effects.

The field now faces critical challenges in translating research findings to clinical practice. Current protocols, requiring extensive therapeutic support in specialized settings, pose scalability challenges. Innovative approaches like group administration show promise for improving accessibility, though their comparative effectiveness remains to be established. Future research must address these pragmatic concerns while maintaining rigorous safety standards and showing durable benefit in diverse populations.

The path forward requires methodological innovation and healthcare system adaptation. Trials incorporating active placebos, assessment of blinding integrity, and systematic documentation of adverse events will strengthen the evidence base.[4] Meanwhile, standardized approaches to delivery, monitoring, and integration support must be developed for safe implementation at scale. Whether these new interventions can fulfil their therapeutic promise while meeting regulatory requirements and healthcare system constraints remains a defining question for the field.

GLOSSARY OF ABBREVIATIONS

- AMPA—α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid
- BDNF—brain derived neurotrophic factor
- CPS-5—clinician administered PTSD scale for DSM-5
- CSTC—cortico-striato-thalamo-cortical
- DMN—default mode network
- DSM-5—Diagnostic and Statistical Manual of Mental Disorders fifth edition
- FDA—United States Food and Drug Administration
- GRID-HAMD—GRID Hamilton depression rating scale
- LSD—lysergic acid diethylamide
- MADRS—Montgomery-Åsberg depression rating scale
- MDMA—3,4-methylenedioxymethamphetamine
- mTOR—mammalian target of rapamycin
- NMDA—N-methyl-D-aspartate receptor activation
- PTSD—post-traumatic stress disorder
- QIDS-SR-16—quick inventory of depressive symptomatology-self-report-16 items
- REBUS—relaxed beliefs under psychedelics
- SSRI—selective serotonin reuptake inhibitor
- TrkB—tropomyosin receptor kinase B

GLOSSARY OF TERMS

- Between-network functional connectivity—degree to which brain regions in different functional networks show temporally correlated activity
- Fear extinction—gradual weakening of a conditioned fear response when the feared stimulus is repeatedly encountered without the expected aversive outcome; the learning process underlying exposure based treatments for post-traumatic stress disorder
- Fear-potentiated startle—enhanced reflexive startle response (eg, eye blink) when a conditioned fear cue is present; a physiological measure of fear memory that is distinct from conscious fear reports
- Functional unblinding—compromise of study blinding that occurs when a, treatment's subjective effects allow the identification of group assignment, potentially inflating apparent efficacy through expectancy effects
- High level prior—in predictive processing theories, a prior is the brain's pre-existing expectation or belief that shapes how incoming sensory information is interpreted. A high level prior refers to abstract beliefs about self, others, or the world (eg, "I am worthless") rather than low level perceptual expectations
- Mystical-type experience—a cluster of often co-occurring subjective phenomena including feelings of unity or interconnectedness, transcendence of time and space, positive mood, and ineffability. Whether these experiences are causally related to therapeutic outcomes remains unclear
- Phenomenological validity—extent to which a self-report measure accurately captures the subjective experience it purports to assess
- Psychological flexibility—capacity to adapt thoughts and behaviors to situational demands while maintaining alignment with personal values; conceptually opposite to experiential avoidance and cognitive rigidity
- Psychological insight—new understanding about oneself, relationships, or existential concerns emerging during or after psychedelic experiences, often accompanied by a subjective sense of clarity or certainty
- Within-network functional connectivity—degree to which brain regions in the same functional network show temporally correlated activity

QUESTIONS FOR FUTURE RESEARCH

- How can trial designs better differentiate direct pharmacological effects, expectancy effects, and psychotherapeutic effects in psychedelic therapy, and what role should functional unblinding play in assessing therapeutic outcomes?
- What are the relative contributions of acute psychological effects (eg, mystical-type experiences, insight) versus sustained biological changes (eg, enhanced neuroplasticity) to therapeutic outcomes?
- What modifications to standard protocols will maintain safety and efficacy while improving scalability after licensing?
- How do therapeutic outcomes and safety profiles differ between the highly screened populations in current trials, and patients with common clinical complexities (eg, psychiatric comorbidities, medication use)?

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1 Yaden DB, Potash JB, Griffiths RR. Preparing for the bursting of the psychedelic hype bubble. *JAMA Psychiatry* 2022;79:943-4. doi:10.1001/jamapsychiatry.2022.2546
2 Lemarchand C, Chopin R, Paul M, et al. Fragile promise of psychedelics in psychiatry. *BMJ* 2024;387:e080391. doi:10.1136/bmj-2024-080391
3 Aday JS, Heifets BD, Pratscher SD, Bradley E, Rosen R, Woolley JD. Great expectations: recommendations for improving the methodological rigor of psychedelic clinical trials. *Psychopharmacology (Berl)* 2022;239:1989-2010. doi:10.1007/s00213-022-06123-7
4 van Elk M, Fried EI. History repeating: guidelines to address common problems in psychedelic science. *Ther Adv Psychopharmacol* 2023;13:20451253231198466. doi:10.1177/20451253231198466
5 Hinkle JT, Graziosi M, Nayak SM, Yaden DB. Adverse events in studies of classic psychedelics: a systematic review and meta-analysis. *JAMA Psychiatry* 2024;81:1225-35. doi:10.1001/jamapsychiatry.2024.2546
6 van Elk M, Yaden DB. Pharmacological, neural, and psychological mechanisms underlying psychedelics: a critical review. *Neurosci Biobehav Rev* 2022;140:104793. doi:10.1016/j.neubiorev.2022.104793
7 Nichols DE. Psychedelics. *Pharmacol Rev* 2016;68:264-355. doi:10.1124/pr.115.011478
8 Kwan AC, Olson DE, Preller KH, Roth BL. The neural basis of psychedelic action. *Nat Neurosci* 2022;25:1407-19. doi:10.1038/s41593-022-01177-4
9 Béïque JC, Campbell B, Perring P, et al. Serotonergic regulation of membrane potential in developing rat prefrontal cortex: coordinated expression of 5-hydroxytryptamine (5-HT)$_{1A}$, 5-HT$_{2A}$, and 5-HT$_{7}$ receptors. *J Neurosci* 2004;24:4807-17. doi:10.1523/JNEUROSCI.5113-03.2004
10 Watakabe A, Komatsu Y, Sadakane O, et al. Enriched expression of serotonin 1B and 2A receptor genes in macaque visual cortex and their bidirectional modulatory effects on neuronal responses. *Cereb Cortex* 2009;19:1915-28. doi:10.1093/cercor/bhn219
11 Beliveau V, Ganz M, Feng L, et al. A high-resolution *in vivo* atlas of the human brain's serotonin system. *J Neurosci* 2017;37:120-8. doi:10.1523/JNEUROSCI.2830-16.2017
12 Madsen MK, Fisher PM, Burmester D, et al. Psychedelic effects of psilocybin correlate with serotonin 2A receptor occupancy and plasma psilocin levels. *Neuropsychopharmacology* 2019;44:1328-34. doi:10.1038/s41386-019-0324-9
13 Preller KH, Herdener M, Pokorny T, et al. The fabric of meaning and subjective effects in LSD-induced states depend on serotonin 2A receptor activation. *Curr Biol* 2017;27:451-7. doi:10.1016/j.cub.2016.12.030
14 Holze F, Vizeli P, Ley L, et al. Acute dose-dependent effects of lysergic acid diethylamide in a double-blind placebo-controlled study in healthy subjects. *Neuropsychopharmacology* 2021;46:537-44. doi:10.1038/s41386-020-00883-6
15 Vollenweider FX, Vollenweider-Scherpenhuyzen MFI, Bäbler A, Vogel H, Hell D. Psilocybin induces schizophrenia-like psychosis in humans via a serotonin-2 agonist action. *Neuroreport* 1998;9:3897-902. doi:10.1097/00001756-199812010-00024
16 González-Maeso J, Weisstaub NV, Zhou M, et al. Hallucinogens recruit specific cortical 5-HT(2A) receptor-mediated signaling pathways to affect behavior. *Neuron* 2007;53:439-52. doi:10.1016/j.neuron.2007.01.008.
17 Wallach J, Cao AB, Calkins MM, et al. Identification of 5-HT$_{2A}$ receptor signaling pathways associated with psychedelic potential. *Nat Commun* 2023;14:8221. doi:10.1038/s41467-023-44016-1
18 Barre A, Berthoux C, De Bundel D, et al. Presynaptic serotonin 2A receptors modulate thalamocortical plasticity and associative learning. *Proc Natl Acad Sci U S A* 2016;113:E1382-91. doi:10.1073/pnas.1525586113
19 Berthoux C, Barre A, Bockaert J, Marin P, Bécamel C. Sustained activation of postsynaptic 5-HT2A receptors gates plasticity at prefrontal cortex synapses. *Cereb Cortex* 2019;29:1659-69. doi:10.1093/cercor/bhy064
20 Szpręgiel I, Bysiek A. Psilocybin and the glutamatergic pathway: implications for the treatment of neuropsychiatric diseases. *Pharmacol Rep* 2024;76:1297-304. doi:10.1007/s43440-024-00660-y
21 Quartarone A, Ghilardi MF, Boller F, editors. *Neuroplasticity: from bench to bedside. Handbook of clinical neurology.* Elsevier, 2022.
22 Noda Y, Zomorrodi R, Vila-Rodriguez F, et al. Impaired neuroplasticity in the prefrontal cortex in depression indexed through paired associative stimulation. *Depress Anxiety* 2018;35:448-56. doi:10.1002/da.22738
23 Appelbaum LG, Shenasa MA, Stolz L, Daskalakis Z. Synaptic plasticity and mental health: methods, challenges and opportunities. *Neuropsychopharmacology* 2023;48:113-20. doi:10.1038/s41386-022-01370-w
24 Jaster AM, de la Fuente Revenga M, González-Maeso J. Molecular targets of psychedelic-induced plasticity. *J Neurochem* 2022;162:80-8. doi:10.1111/jnc.15536
25 Ignácio ZM, Réus GZ, Arent CO, Abelaira HM, Pitcher MR, Quevedo J. New perspectives on the involvement of mTOR in depression as well as in the action of antidepressant drugs. *Br J Clin Pharmacol* 2016;82:1280-90. doi:10.1111/bcp.12845
26 Ly C, Greb AC, Cameron LP, et al. Psychedelics promote structural and functional neural plasticity. *Cell Rep* 2018;23:3170-82. doi:10.1016/j.celrep.2018.05.022
27 Moliner R, Girych M, Brunello CA, et al. Psychedelics promote plasticity by directly binding to BDNF receptor TrkB. *Nat Neurosci* 2023;26:1032-41. doi:10.1038/s41593-023-01316-5
28 Calder AE, Hase A, Hasler G. Effects of psychoplastogens on blood levels of brain-derived neurotrophic factor (BDNF) in humans: a systematic review and meta-analysis. *Mol Psychiatry* 2025;30:763-76. doi:10.1038/s41380-024-02830-z
29 Shao LX, Liao C, Gregg I, et al. Psilocybin induces rapid and persistent growth of dendritic spines in frontal cortex in vivo. *Neuron* 2021;109:2535-2544.e4. doi:10.1016/j.neuron.2021.06.008
30 Xu T, Yu X, Perlik AJ, et al. Rapid formation and selective stabilization of synapses for enduring motor memories. *Nature* 2009;462:915-9. doi:10.1038/nature08389
31 Raval NR, Johansen A, Donovan LL, et al. A single dose of psilocybin increases synaptic density and decreases 5-HT$_{2A}$ receptor density in the pig brain. *Int J Mol Sci* 2021;22:835. doi:10.3390/ijms22020835
32 Abraham WC, Bear MF. Metaplasticity: the plasticity of synaptic plasticity. *Trends Neurosci* 1996;19:126-30. doi:10.1016/S0166-2236(96)80018-X
33 de la Fuente Revenga M, Zhu B, Guevara CA, et al. Prolonged epigenomic and synaptic plasticity alterations following single exposure to a psychedelic in mice. *Cell Rep* 2021;37:109836. doi:10.1016/j.celrep.2021.109836
34 Zahid Z, Sultan ZW, Krause BM, Wenthur CJ, Pearce RA, Banks MI. Divergent effects of ketamine and the serotoninergic psychedelic 2,5-dimethoxy-4-iodoamphetamine on hippocampal plasticity and metaplasticity. *Psychedelic Med (New Rochelle)* 2024;2:166-77. doi:10.1089/psymed.2023.0061
35 Morris L, Mansell W. A systematic review of the relationship between rigidity/flexibility and transdiagnostic cognitive and behavioral processes that maintain psychopathology. *J Exp Psychopathol* 2018;9:2043808718779431. doi:10.1177/2043808718779431 .
36 Sheppes G, Suri G, Gross JJ. Emotion regulation and psychopathology. *Annu Rev Clin Psychol* 2015;11:379-405. doi:10.1146/annurev-clinpsy-032814-112739
37 McCulloch DEW, Knudsen GM, Barrett FS, et al. Psychedelic resting-state neuroimaging: a review and perspective on balancing replication and novel analyses. *Neurosci Biobehav Rev* 2022;138:104689. doi:10.1016/j.neubiorev.2022.104689
38 Shinozuka K, Jerotic K, Mediano P, et al. Synergistic, multi-level understanding of psychedelics: three systematic reviews and meta-analyses of their pharmacology, neuroimaging and phenomenology. *Transl Psychiatry* 2024;14:485. doi:10.1038/s41398-024-03187-1
39 Preller KH, Duerler P, Burt JB, et al. Psilocybin induces time-dependent changes in global functional connectivity. *Biol Psychiatry* 2020;88:197-207. doi:10.1016/j.biopsych.2019.12.027
40 Copa D, Erritzoe D, Giribaldi B, Nutt D, Carhart-Harris R, Tagliazucchi E. Predicting the outcome of psilocybin treatment for depression from baseline fMRI functional connectivity. *J Affect Disord* 2024;353:60-9. doi:10.1016/j.jad.2024.02.089
41 Menon V. 20 years of the default mode network: a review and synthesis. *Neuron* 2023;111:2469-87. doi:10.1016/j.neuron.2023.04.023
42 Doucet GE, Janiri D, Howard R, O'Brien M, Andrews-Hanna JR, Frangou S. Transdiagnostic and disease-specific abnormalities in the default-mode network hubs in psychiatric disorders: a

BMJ: first published as 10.1136/bmj-2024-081723 on 23 February 2026. Downloaded from https://www.bmj.com/ on 15 March 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

meta-analysis of resting-state functional imaging studies. *Eur Psychiatry* 2020;63:e57. doi:10.1192/j.eurpsy.2020.57
43 Gattuso JJ, Perkins D, Ruffell S, et al. Default mode network modulation by psychedelics: a systematic review. *Int J Neuropsychopharmacol* 2023;26:155-88. doi:10.1093/ijnp/pyac074
44 Weber AM, Soreni N, Noseworthy MD. A preliminary study on the effects of acute ethanol ingestion on default mode network and temporal fractal properties of the brain. *MAGMA* 2014;27:291-301. doi:10.1007/s10334-013-0420-5
45 Schrantee A, Ferguson B, Stoffers D, Booij J, Rombouts S, Reneman L. Effects of dexamphetamine-induced dopamine release on resting-state network connectivity in recreational amphetamine users and healthy controls. *Brain Imaging Behav* 2016;10:548-58. doi:10.1007/s11682-015-9419-z
46 Klaassens BL, van Gorsel HC, Khalili-Mahani N, et al. Single-dose serotonergic stimulation shows widespread effects on functional brain connectivity. *Neuroimage* 2015;122:440-50. doi:10.1016/j.neuroimage.2015.08.012
47 Choi KW, Han KM, Kim H, et al. Comparison of shape alterations of the thalamus and caudate nucleus between drug-naïve major depressive disorder patients and healthy controls. *J Affect Disord* 2020;264:279-85. doi:10.1016/j.jad.2020.01.011
48 Li B, Mody M. Cortico-striato-thalamo-cortical circuitry, working memory, and obsessive-compulsive disorder. *Front Psychiatry* 2016;7:78. doi:10.3389/fpsyt.2016.00078
49 Feil J, Sheppard D, Fitzgerald PB, Yücel M, Lubman DI, Bradshaw JL. Addiction, compulsive drug seeking, and the role of frontostriatal mechanisms in regulating inhibitory control. *Neurosci Biobehav Rev* 2010;35:248-75. doi:10.1016/j.neubiorev.2010.03.001
50 Mueller F, Lenz C, Dolder PC, et al. Acute effects of LSD on amygdala activity during processing of fearful stimuli in healthy subjects. *Transl Psychiatry* 2017;7:e1084. doi:10.1038/tp.2017.54
51 Preller KH, Burt JB, Ji JL, et al. Changes in global and thalamic brain connectivity in LSD-induced altered states of consciousness are attributable to the 5-HT2A receptor. *Elife* 2018;7:e35082. doi:10.7554/eLife.35082
52 Quednow BB, Kometer M, Geyer MA, Vollenweider FX. Psilocybin-induced deficits in automatic and controlled inhibition are attenuated by ketanserin in healthy human volunteers. *Neuropsychopharmacology* 2012;37:630-40. doi:10.1038/npp.2011.228
53 Schmid Y, Enzler F, Gasser P, et al. Acute effects of lysergic acid diethylamide in healthy subjects. *Biol Psychiatry* 2015;78:544-53. doi:10.1016/j.biopsych.2014.11.015
54 Vollenweider FX, Csomor PA, Knappe B, Geyer MA, Quednow BB. The effects of the preferential 5-HT2A agonist psilocybin on prepulse inhibition of startle in healthy human volunteers depend on interstimulus interval. *Neuropsychopharmacology* 2007;32:1876-87. doi:10.1038/sj.npp.1301324
55 Barrett FS, Doss MK, Sepeda ND, Pekar JJ, Griffiths RR. Emotions and brain function are altered up to one month after a single high dose of psilocybin. *Sci Rep* 2020;10:2214. doi:10.1038/s41598-020-59282-y
56 Stroud JB, Freeman TP, Leech R, et al. Psilocybin with psychological support improves emotional face recognition in treatment-resistant depression. *Psychopharmacology (Berl)* 2018;235:459-66. doi:10.1007/s00213-017-4754-y
57 Roseman L, Demetriou L, Wall MB, Nutt DJ, Carhart-Harris RL. Increased amygdala responses to emotional faces after psilocybin for treatment-resistant depression. *Neuropharmacology* 2018;142:263-9. doi:10.1016/j.neuropharm.2017.12.041
58 Zeifman RJ, Wagner AC, Watts R, Kettner H, Mertens LJ, Carhart-Harris RL. Post-psychedelic reductions in experiential avoidance are associated with decreases in depression severity and suicidal ideation. *Front Psychiatry* 2020;11:782. doi:10.3389/fpsyt.2020.00782
59 Zeifman RJ, Wagner AC, Monson CM, Carhart-Harris RL. How does psilocybin therapy work? An exploration of experiential avoidance as a putative mechanism of change. *J Affect Disord* 2023;334:100-12. doi:10.1016/j.jad.2023.04.105
60 Carhart-Harris RL, Friston KJ. REBUS and the anarchic brain: toward a unified model of the brain action of psychedelics. *Pharmacol Rev* 2019;71:316-44. doi:10.1124/pr.118.017160
61 Timmermann C, Spriggs MJ, Kaelen M, et al. LSD modulates effective connectivity and neural adaptation mechanisms in an auditory oddball paradigm. *Neuropharmacology* 2018;142:251-62. doi:10.1016/j.neuropharm.2017.10.039
62 Siegel JS, Subramanian S, Perry D, et al. Psilocybin desynchronizes the human brain. *Nature* 2024;632:131-8. doi:10.1038/s41586-024-07624-5
63 Doss MK, Považan M, Rosenberg MD, et al. Psilocybin therapy increases cognitive and neural flexibility in patients with major depressive disorder. *Transl Psychiatry* 2021;11:574. doi:10.1038/s41398-021-01706-y
64 Nayak SM, Jackson H, Sepeda ND, et al. Naturalistic psilocybin use is associated with persisting improvements in mental health and wellbeing: results from a prospective, longitudinal survey. *Front Psychiatry* 2023;14:1199642. doi:10.3389/fpsyt.2023.1199642
65 Fisher EL, Smith R, Conn K, et al. Psilocybin increases optimistic engagement over time: computational modelling of behaviour in rats. *Transl Psychiatry* 2024;14:394. doi:10.1038/s41398-024-03103-7
66 Yaden DB, Griffiths RR. The subjective effects of psychedelics are necessary for their enduring therapeutic effects. *ACS Pharmacol Transl Sci* 2020;4:568-72. doi:10.1021/acsptsci.0c00194
67 Olson DE. The subjective effects of psychedelics may not be necessary for their enduring therapeutic effects. *ACS Pharmacol Transl Sci* 2020;4:563-7. doi:10.1021/acsptsci.0c00192
68 Ross S, Bossis A, Guss J, et al. Rapid and sustained symptom reduction following psilocybin treatment for anxiety and depression in patients with life-threatening cancer: a randomized controlled trial. *J Psychopharmacol* 2016;30:1165-80. doi:10.1177/0269881116675512
69 Griffiths RR, Johnson MW, Carducci MA, et al. Psilocybin produces substantial and sustained decreases in depression and anxiety in patients with life-threatening cancer: a randomized double-blind trial. *J Psychopharmacol* 2016;30:1181-97. doi:10.1177/0269881116675513
70 Roseman L, Nutt DJ, Carhart-Harris RL. Quality of acute psychedelic experience predicts therapeutic efficacy of psilocybin for treatment-resistant depression. *Front Pharmacol* 2018;8:974. doi:10.3389/fphar.2017.00974
71 Garcia-Romeu A, Griffiths RR, Johnson MW. Psilocybin-occasioned mystical experiences in the treatment of tobacco addiction. *Curr Drug Abuse Rev* 2014;7:157-64. doi:10.2174/1874473708666150107121331
72 Bogenschutz MP, Forcehimes AA, Pommy JA, Wilcox CE, Barbosa PC, Strassman RJ. Psilocybin-assisted treatment for alcohol dependence: a proof-of-concept study. *J Psychopharmacol* 2015;29:289-99. doi:10.1177/0269881114565144
73 Yaden DB, Goldy SP, Weiss B, Griffiths RR. Clinically relevant acute subjective effects of psychedelics beyond mystical experience. *Nat Rev Psychol* 2024;3:606-21. doi:10.1038/s44159-024-00345-6
74 Nautiyal KM, Yaden DB. Does the trip matter? Investigating the role of the subjective effects of psychedelics in persisting therapeutic effects. *Neuropsychopharmacology* 2023;48:215-6. doi:10.1038/s41386-022-01424-z
75 Roseman L, Haijen E, Idialu-Ikato K, Kaelen M, Watts R, Carhart-Harris R. Emotional breakthrough and psychedelics: validation of the Emotional Breakthrough Inventory. *J Psychopharmacol* 2019;33:1076-87. doi:10.1177/0269881119855974
76 Goodwin GM, Aaronson ST, Alvarez O, et al. The role of the psychedelic experience in psilocybin treatment for treatment-resistant depression. *J Affect Disord* 2025;372:523-32. doi:10.1016/j.jad.2024.12.061
77 Kugel J, Laukkonen RE, Yaden DB, Yücel M, Liknaitzky P. Insights on psychedelics: a systematic review of therapeutic effects. *Neurosci Biobehav Rev* 2025;173:106117. doi:10.1016/j.neubiorev.2025.106117
78 Davis AK, Barrett FS, Griffiths RR. Psychological flexibility mediates the relations between acute psychedelic effects and subjective decreases in depression and anxiety. *J Contextual Behav Sci* 2020;15:39-45. doi:10.1016/j.jcbs.2019.11.004
79 Davis AK, Xin Y, Sepeda ND, Garcia-Romeu A, Williams MT. Increases in psychological flexibility mediate relationship between acute psychedelic effects and decreases in racial trauma symptoms among people of color. *Chronic Stress (Thousand Oaks)* 2021;5:24705470211035607. doi:10.1177/24705470211035607
80 Vollenweider FX, Liechti ME, Gamma A, Greer G, Geyer M. Acute psychological and neurophysiological effects of MDMA in humans. *J Psychoactive Drugs* 2002;34:171-84. doi:10.1080/02791072.2002.10399951
81 Liechti ME, Vollenweider FX. Which neuroreceptors mediate the subjective effects of MDMA in humans? A summary of mechanistic studies. *Hum Psychopharmacol* 2001;16:589-98. doi:10.1002/hup.348
82 Carhart-Harris RL, Murphy K, Leech R, et al. The effects of acutely administered 3,4-methylenedioxymethamphetamine on spontaneous brain function in healthy volunteers measured with arterial spin labeling and blood oxygen level-dependent resting state functional connectivity. *Biol Psychiatry* 2015;78:554-62. doi:10.1016/j.biopsych.2013.12.015
83 Qasim SE, Mohan UR, Stein JM, Jacobs J. Neuronal activity in the human amygdala and hippocampus enhances emotional memory encoding. *Nat Hum Behav* 2023;7:754-64. doi:10.1038/s41562-022-01502-8

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84 Ressler KJ. Amygdala activity, fear, and anxiety: modulation by stress. *Biol Psychiatry* 2010;67:1117-9. doi:10.1016/j.biopsych.2010.04.027
85 Iqbal J, Huang GD, Xue YX, Yang M, Jia XJ. The neural circuits and molecular mechanisms underlying fear dysregulation in posttraumatic stress disorder. *Front Neurosci* 2023;17:1281401. doi:10.3389/fnins.2023.1281401
86 Sripada RK, King AP, Garfinkel SN, et al. Altered resting-state amygdala functional connectivity in men with posttraumatic stress disorder. *J Psychiatry Neurosci* 2012;37:241-9. doi:10.1503/jpn.110069
87 Singleton SP, Wang JB, Mithoefer M, et al. Altered brain activity and functional connectivity after MDMA-assisted therapy for post-traumatic stress disorder. *Front Psychiatry* 2023;13:947622. doi:10.3389/fpsyt.2022.947622
88 Müller F, Holze F, Dolder P, et al. MDMA-induced changes in within-network connectivity contradict the specificity of these alterations for the effects of serotonergic hallucinogens. *Neuropsychopharmacology* 2021;46:545-53. doi:10.1038/s41386-020-00906-2
89 Roseman L, Leech R, Feilding A, Nutt DJ, Carhart-Harris RL. The effects of psilocybin and MDMA on between-network resting state functional connectivity in healthy volunteers. *Front Hum Neurosci* 2014;8:204. doi:10.3389/fnhum.2014.00204
90 Nardou R, Lewis EM, Rothhaas R, et al. Oxytocin-dependent reopening of a social reward learning critical period with MDMA. *Nature* 2019;569:116-20. doi:10.1038/s41586-019-1075-9
91 Nardou R, Sawyer E, Song YJ, et al. Psychedelics reopen the social reward learning critical period. *Nature* 2023;618:790-8. doi:10.1038/s41586-023-06204-3
92 Lu OD, White K, Raymond K, et al. (2025). A multi-institutional investigation of psilocybin's effects on mouse behavior. bioRxiv, 2025-04.
93 Grinspoon L, Bakalar JB. *Psychedelic drugs reconsidered.* Basic Books, 1979.
94 Schmid Y, Hysek CM, Simmler LD, Crockett MJ, Quednow BB, Liechti ME. Differential effects of MDMA and methylphenidate on social cognition. *J Psychopharmacol* 2014;28:847-56. doi:10.1177/0269881114542454
95 Wardle MC, Kirkpatrick MG, de Wit H. "Ecstasy" as a social drug: MDMA preferentially affects responses to emotional stimuli with social content. *Soc Cogn Affect Neurosci* 2014;9:1076-81. doi:10.1093/scan/nsu035
96 Hysek CM, Schmid Y, Simmler LD, et al. MDMA enhances emotional empathy and prosocial behavior. *Soc Cogn Affect Neurosci* 2014;9:1645-52. doi:10.1093/scan/nst161
97 Stewart LH, Ferguson B, Morgan CJ, et al. Effects of ecstasy on cooperative behaviour and perception of trustworthiness: a naturalistic study. *J Psychopharmacol* 2014;28:1001-8. doi:10.1177/0269881114544775
98 Bedi G, Hyman D, de Wit H. Is ecstasy an "empathogen"? Effects of ±3,4-methylenedioxymethamphetamine on prosocial feelings and identification of emotional states in others. *Biol Psychiatry* 2010;68:1134-40. doi:10.1016/j.biopsych.2010.08.003
99 Baggott MJ, Coyle JR, Siegrist JD, Garrison KJ, Galloway GP, Mendelson JE. Effects of 3,4-methylenedioxymethamphetamine on socioemotional feelings, authenticity, and autobiographical disclosure in healthy volunteers in a controlled setting. *J Psychopharmacol* 2016;30:378-87. doi:10.1177/0269881115626348
100 Kirkpatrick MG, Francis SM, Lee R, de Wit H, Jacob S. Plasma oxytocin concentrations following MDMA or intranasal oxytocin in humans. *Psychoneuroendocrinology* 2014;46:23-31. doi:10.1016/j.psyneuen.2014.04.006
101 Atila C, Holze F, Murugesu R, et al. Oxytocin in response to MDMA provocation test in patients with arginine vasopressin deficiency (central diabetes insipidus): a single-centre, case-control study with nested, randomised, double-blind, placebo-controlled crossover trial. *Lancet Diabetes Endocrinol* 2023;11:454-64. doi:10.1016/S2213-8587(23)00120-1
102 O'Donnell KC, Okano L, Alpert M, et al. The conceptual framework for the therapeutic approach used in phase 3 trials of MDMA-assisted therapy for PTSD. *Front Psychol* 2024;15:1427531. doi:10.3389/fpsyg.2024.1427531
103 Schrader C, Ross A. A review of PTSD and current treatment strategies. *Mo Med* 2021;118:546-51.
104 Rauch SAM, Eftekhari A, Ruzek JI. Review of exposure therapy: a gold standard for PTSD treatment. *J Rehabil Res Dev* 2012;49:679-87. doi:10.1682/JRRD.2011.08.0152
105 Young MB, Andero R, Ressler KJ, Howell LL. 3,4-Methylenedioxymethamphetamine facilitates fear extinction learning. *Transl Psychiatry* 2015;5:e634. doi:10.1038/tp.2015.138
106 Hake HS, Davis JKP, Wood RR, et al. 3,4-Methylenedioxymethamphetamine (MDMA) impairs the extinction and reconsolidation of fear memory in rats. *Physiol Behav* 2019;199:343-50. doi:10.1016/j.physbeh.2018.12.007
107 Young MB, Norrholm SD, Khoury LM, et al. Inhibition of serotonin transporters disrupts the enhancement of fear memory extinction by 3,4-methylenedioxymethamphetamine (MDMA). *Psychopharmacology (Berl)* 2017;234:2883-95. doi:10.1007/s00213-017-4684-8
108 Wicking M, Steiger F, Nees F, et al. Deficient fear extinction memory in posttraumatic stress disorder. *Neurobiol Learn Mem* 2016;136:116-26. doi:10.1016/j.nlm.2016.09.016
109 Vizeli P, Straumann I, Duthaler U, et al. Effects of 3,4-methylenedioxymethamphetamine on conditioned fear extinction and retention in a crossover study in healthy subjects. *Front Pharmacol* 2022;13:906639. doi:10.3389/fphar.2022.906639
110 Ronconi JM, Shiner B, Watts BV. Inclusion and exclusion criteria in randomized controlled trials of psychotherapy for PTSD. *J Psychiatr Pract* 2014;20:25-37. doi:10.1097/01.pra.0000442936.23457.5b
111 Smith SM, Goldstein RB, Grant BF. The association between post-traumatic stress disorder and lifetime DSM-5 psychiatric disorders among veterans: data from the National Epidemiologic Survey on Alcohol and Related Conditions-III (NESARC-III). *J Psychiatr Res* 2016;82:16-22. doi:10.1016/j.jpsychires.2016.06.022
112 Pietrzak RH, Goldstein RB, Southwick SM, Grant BF. Prevalence and axis I comorbidity of full and partial posttraumatic stress disorder in the United States: results from Wave 2 of the National Epidemiologic Survey on Alcohol and Related Conditions. *J Anxiety Disord* 2011;25:456-65. doi:10.1016/j.janxdis.2010.11.010
113 Mitchell JM, Bogenschutz M, Lilienstein A, et al. MDMA-assisted therapy for severe PTSD: a randomized, double-blind, placebo-controlled phase 3 study. *Nat Med* 2021;27:1025-33. doi:10.1038/s41591-021-01336-3
114 Mitchell JM, Ot'alora G M, van der Kolk B, et al, MAPP2 Study Collaborator Group. MDMA-assisted therapy for moderate to severe PTSD: a randomized, placebo-controlled phase 3 trial. *Nat Med* 2023;29:2473-80. doi:10.1038/s41591-023-02565-4
115 Yang J, Wang N, Luo W, Gao J. The efficacy and safety of MDMA-assisted psychotherapy for treatment of posttraumatic stress disorder: a systematic review and meta-analysis from randomized controlled trials. *Psychiatry Res* 2024;339:116043. doi:10.1016/j.psychres.2024.116043
116 Hróbjartsson A, Emanuelsson F, Skou Thomsen AS, Hilden J, Brorson S. Bias due to lack of patient blinding in clinical trials. A systematic review of trials randomizing patients to blind and nonblind sub-studies. *Int J Epidemiol* 2014;43:1272-83. doi:10.1093/ije/dyu115
117 Fergusson D, Glass KC, Waring D, Shapiro S. Turning a blind eye: the success of blinding reported in a random sample of randomised, placebo controlled trials. *BMJ* 2004;328:432.
118 Kelmendi B, Adams TG, Yarnell S, Southwick S, Abdallah CG, Krystal JH. PTSD: from neurobiology to pharmacological treatments. *Eur J Psychotraumatol* 2016;7:31858. doi:10.3402/ejpt.v7.31858
119 Bschor T, Nagel L, Unger J, Schwarzer G, Baethge C. Differential outcomes of placebo treatment across 9 psychiatric disorders: a systematic review and meta-analysis. *JAMA Psychiatry* 2024;81:757-68. doi:10.1001/jamapsychiatry.2024.0994
120 Szigeti B, Bradley ER, Woolley J. Too big to fail? Comparing effect sizes of MDMA assisted therapy to unmasking bias. *Psychopharmacology (Berl)* 2025. doi:10.1007/s00213-025-06886-9
121 Mustafa R, McQueen B, Nikitin D, Nhan E, Zempleyeni A. 3,4-Methylenedioxymethamphetamine-assisted psychotherapy for post-traumatic stress disorder: effectiveness and value. Institute for Clinical and Economic Review; 2024 (cited 10 Jan 2025). Available from: https://icer.org/wp-content/uploads/2023/11/PTSD_Revised-Report_For-Publication_05142024.pdf
122 Lykos Therapeutics. Psychopharmacologic Drugs Advisory Committee briefing presentation: MDMA-assisted therapy for PTSD. Psychopharmacologic Drugs Advisory Committee (PDAC) meeting (FDA); 4 Jun 2024.
123 US Food and Drug Administration. Psychopharmacologic Drugs Advisory Committee meeting: Midomafetamine (MDMA) capsules (NDA 215455) [transcript]. pp 98–100. Available from: https://www.fda.gov/media/180703/download
124 Davis AK, Barrett FS, May DG, et al. Effects of psilocybin-assisted therapy on major depressive disorder: a randomized clinical trial. *JAMA Psychiatry* 2021;78:481-9. doi:10.1001/jamapsychiatry.2020.3285
125 Gukasyan N, Davis AK, Barrett FS, et al. Efficacy and safety of psilocybin-assisted treatment for major depressive disorder: prospective 12-month follow-up. *J Psychopharmacol* 2022;36:151-8. doi:10.1177/02698811211073759
126 Raison CL, Sanacora G, Woolley J, et al. Single-dose psilocybin treatment for major depressive disorder: a randomized clinical trial. *JAMA* 2023;330:843-53. doi:10.1001/jama.2023.14530

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127 Carhart-Harris R, Giribaldi B, Watts R, et al. Trial of psilocybin versus escitalopram for depression. *N Engl J Med* 2021;384:1402-11. doi:10.1056/NEJMoa2032994
128 Erritzoe D, Barba T, Greenway KT, et al. Effect of psilocybin versus escitalopram on depression symptom severity in patients with moderate-to-severe major depressive disorder: observational 6-month follow-up of a phase 2, double-blind, randomised, controlled trial. *EClinicalMedicine* 2024;76:102799. doi:10.1016/j.eclinm.2024.102799
129 Goodwin GM, Aaronson ST, Alvarez O, et al. Single-dose psilocybin for a treatment-resistant episode of major depression. *N Engl J Med* 2022;387:1637-48. doi:10.1056/NEJMoa2206443
130 Goodwin GM, Aaronson ST, Alvarez O, et al. Single-dose psilocybin for a treatment-resistant episode of major depression: impact on patient-reported depression severity, anxiety, function, and quality of life. *J Affect Disord* 2023;327:120-7. doi:10.1016/j.jad.2023.01.108
131 Rosenblat JD, Meshkat S, Doyle Z, et al. Psilocybin-assisted psychotherapy for treatment resistant depression: a randomized clinical trial evaluating repeated doses of psilocybin. *Med* 2024;5:190-200.e5. doi:10.1016/j.medj.2024.01.005
132 Yaden DB, Berghella AP, Hendricks PS, et al. IUPHAR-review: the integration of classic psychedelics into current substance use disorder treatment models. *Pharmacol Res* 2024;199:106998. doi:10.1016/j.phrs.2023.106998
133 Krebs TS, Johansen PØ. Lysergic acid diethylamide (LSD) for alcoholism: meta-analysis of randomized controlled trials. *J Psychopharmacol* 2012;26:994-1002. doi:10.1177/0269881112439253
134 Zafar R, Siegel M, Harding R, et al. Psychedelic therapy in the treatment of addiction: the past, present and future. *Front Psychiatry* 2023;14:1183740. doi:10.3389/fpsyt.2023.1183740
135 Bogenschutz MP, Ross S, Bhatt S, et al. Percentage of heavy drinking days following psilocybin-assisted psychotherapy vs placebo in the treatment of adult patients with alcohol use disorder: a randomized clinical trial. *JAMA Psychiatry* 2022;79:953-62. doi:10.1001/jamapsychiatry.2022.2096
136 Luquiens A, Belahda D, Graux C, et al. Psilocybin in alcohol use disorder and comorbid depressive symptoms: Results from a feasibility randomized clinical trial. *Addiction* 2025. doi:10.1111/add.70152
137 Rieser NM, Bitar R, Halm S, et al. Psilocybin-assisted therapy for relapse prevention in alcohol use disorder: a phase 2 randomized clinical trial. *EClinicalMedicine* 2025;82:103149. doi:10.1016/j.eclinm.2025.103149
138 Johnson MW, Garcia-Romeu A, Cosimano MP, Griffiths RR. Pilot study of the 5-HT_{2A}R agonist psilocybin in the treatment of tobacco addiction. *J Psychopharmacol* 2014;28:983-92. doi:10.1177/0269881114548296
139 Johnson MW, Garcia-Romeu A, Griffiths RR. Long-term follow-up of psilocybin-facilitated smoking cessation. *Am J Drug Alcohol Abuse* 2017;43:55-60. doi:10.3109/00952990.2016.1170135
140 Schimmers N, Breeksema JJ, Smith-Apeldoorn SY, Veraart J, van den Brink W, Schoevers RA. Psychedelics for the treatment of depression, anxiety, and existential distress in patients with a terminal illness: a systematic review. *Psychopharmacology (Berl)* 2022;239:15-33. doi:10.1007/s00213-021-06027-y
141 Maia LO, Beaussant Y, Garcia ACM. The therapeutic potential of psychedelic-assisted therapies for symptom control in patients diagnosed with serious illness: a systematic review. *J Pain Symptom Manage* 2022;63:e725-38. doi:10.1016/j.jpainsymman.2022.01.024
142 Gasser P, Holstein D, Michel Y, et al. Safety and efficacy of lysergic acid diethylamide-assisted psychotherapy for anxiety associated with life-threatening diseases. *J Nerv Ment Dis* 2014;202:513-20. doi:10.1097/NMD.0000000000000113
143 Grob CS, Danforth AL, Chopra GS, et al. Pilot study of psilocybin treatment for anxiety in patients with advanced-stage cancer. *Arch Gen Psychiatry* 2011;68:71-8. doi:10.1001/archgenpsychiatry.2010.116
144 Agin-Liebes GI, Malone T, Yalch MM, et al. Long-term follow-up of psilocybin-assisted psychotherapy for psychiatric and existential distress in patients with life-threatening cancer. *J Psychopharmacol* 2020;34:155-66. doi:10.1177/0269881119897615
145 Colcott J, Guerin AA, Carter O, Meikle S, Bedi G. Side-effects of MDMA-assisted psychotherapy: a systematic review and meta-analysis. *Neuropsychopharmacology* 2024;49:1208-26. doi:10.1038/s41386-024-01865-8
146 Johnson M, Richards W, Griffiths R. Human hallucinogen research: guidelines for safety. *J Psychopharmacol* 2008;22:603-20. doi:10.1177/0269881108093587
147 Schlag AK, Aday J, Salam I, Neill JC, Nutt DJ. Adverse effects of psychedelics: from anecdotes and misinformation to systematic science. *J Psychopharmacol* 2022;36:258-72. doi:10.1177/02698811211069100
148 Yerubandi A, Thomas JE, Bhuiya NMMA, Harrington C, Villa Zapata L, Caballero J. Acute adverse effects of therapeutic doses of psilocybin: a systematic review and meta-analysis. *JAMA Netw Open* 2024;7:e245960. doi:10.1001/jamanetworkopen.2024.5960
149 Romeo B, Kervadec E, Fauvel B, et al. Safety and risk assessment of psychedelic psychotherapy: a meta-analysis and systematic review. *Psychiatry Res* 2024;335:115880. doi:10.1016/j.psychres.2024.115880
150 Kaminski D, Reinert JP. The tolerability and safety of psilocybin in psychiatric and substance-dependence conditions: a systematic review. *Ann Pharmacother* 2024;58:811-26. doi:10.1177/10600280231205645
151 Sabé M, Sulstarova A, Glangetas A, et al. Reconsidering evidence for psychedelic-induced psychosis: an overview of reviews, a systematic review, and meta-analysis of human studies. *Mol Psychiatry* 2025;30:1223-55. doi:10.1038/s41380-024-02800-5
152 Simonsson O, Goldberg SB, Osika W, et al. Longitudinal associations of naturalistic psychedelic use with psychotic and manic symptoms. *Psychol Med* 2025;55:e99. doi:10.1017/S0033291725000716
153 Honk L, Stenfors CUD, Goldberg SB, et al. Longitudinal associations between psychedelic use and psychotic symptoms in the United States and the United Kingdom. *J Affect Disord* 2024;351:194-201. doi:10.1016/j.jad.2024.01.197
154 Dourron HM, Bradley M, Simonsson O, et al. Naturalistic psychedelic use and psychotic symptoms: a cross-sectional study of individuals with a personal or family history of psychotic or bipolar disorders. *Psychedelic Med (New Rochelle)* 2025;3:93-102. doi:10.1089/psymed.2024.0036
155 Becker AM, Holze F, Grandinetti T, et al. Acute effects of psilocybin after escitalopram or placebo pretreatment in a randomized, double-blind, placebo-controlled, crossover study in healthy subjects. *Clin Pharmacol Ther* 2022;111:886-95. doi:10.1002/cpt.2487
156 Goodwin GM, Croal M, Feifel D, et al. Psilocybin for treatment resistant depression in patients taking a concomitant SSRI medication. *Neuropsychopharmacology* 2023;48:1492-9. doi:10.1038/s41386-023-01648-7
157 Nayak SM, Gukasyan N, Barrett FS, Erowid E, Erowid F, Griffiths RR. Classic psychedelic coadministration with lithium, but not lamotrigine, is associated with seizures: an analysis of online psychedelic experience reports. *Pharmacopsychiatry* 2021;54:240-5. doi:10.1055/a-1524-2794
158 Sarparast A, Thomas K, Malcolm B, Stauffer CS. Drug-drug interactions between psychiatric medications and MDMA or psilocybin: a systematic review. *Psychopharmacology (Berl)* 2022;239:1945-76. doi:10.1007/s00213-022-06083-y
159 Liechti ME, Geyer MA, Hell D, Vollenweider FX. Effects of MDMA (ecstasy) on prepulse inhibition and habituation of startle in humans after pretreatment with citalopram, haloperidol, or ketanserin. *Neuropsychopharmacology* 2001;24:240-52. doi:10.1016/S0893-133X(00)00199-8
160 Junqueira DR, Zorzela L, Golder S, et al, CONSORT Harms Group. CONSORT Harms 2022 statement, explanation, and elaboration: updated guideline for the reporting of harms in randomised trials. *BMJ* 2023;381:e073725. doi:10.1136/bmj-2022-073725
161 Papanikolaou PN, Ioannidis JPA. Availability of large-scale evidence on specific harms from systematic reviews of randomized trials. *Am J Med* 2004;117:582-9. doi:10.1016/j.amjmed.2004.04.026
162 Meister R, von Wolff A, Mohr H, et al. Adverse event methods were heterogeneous and insufficiently reported in randomized trials on persistent depressive disorder. *J Clin Epidemiol* 2016;71:97-108. doi:10.1016/j.jclinepi.2015.10.007
163 Anderson RM, Peña A, Magee T, et al. Reporting of harms in randomized controlled trials published in urology journals: an updated analysis. *J Urol* 2024;211:48-54. doi:10.1097/JU.0000000000003740
164 Ehrenkranz R, Agrawal M, Nayak SM, Yaden DB. Adverse events should not be surprising in psychedelic research. *Psychedelic Med (New Rochelle)* 2025;3:59-62. doi:10.1089/psymed.2024.0006
165 US Food and Drug Administration. Psychedelic drugs: considerations for clinical investigations guidance for industry. US Department of Health and Human Services, Food and Drug Administration, Center for Drug Evaluation and Research (CDER); 2023 (cited 17 Dec 2024). Available from: https://www.fda.gov/media/169694/download
166 Compass Pathways. Compass Pathways successfully achieves primary endpoint in first phase 3 trial evaluating COMP360 psilocybin for treatment-resistant depression. 2025 (cited 25 Aug 2025). Available from: https://ir.compasspathways.com/News--Events-/news/news-details/2025/Compass-Pathways-Successfully-Achieves-Primary-Endpoint-in-First-Phase-3-Trial-Evaluating-COMP360-Psilocybin-for-Treatment-Resistant-Depression/default.aspx

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167 Dutcher EG, Krystal AD. Treatment expectancies and psilocybin vs escitalopram for depression. *JAMA Psychiatry* 2025;82:321-2. doi:10.1001/jamapsychiatry.2024.4387
168 Holze F, Gasser P, Müller F, Dolder PC, Liechti ME. Lysergic acid diethylamide-assisted therapy in patients with anxiety with and without a life-threatening illness: a randomized, double-blind, placebo-controlled phase II study. *Biol Psychiatry* 2023;93:215-23. doi:10.1016/j.biopsych.2022.08.025
169 Beckley Psytech. Positive data from phase 2a study of BPL-003 with SSRIs for treatment-resistant depression. 2025. https://ir.atai.com/news-releases/news-release-details/atai-life-sciences-and-beckley-psytech-report-positive-phase-2a
170 Agrawal M, Emanuel E, Richards B, Richards W, Roddy K, Thambi P. Assessment of psilocybin therapy for patients with cancer and major depression disorder. *JAMA Oncol* 2023;9:864-6. doi:10.1001/jamaoncol.2023.0351
171 Lewis BR, Garland EL, Byrne K, et al. HOPE: a pilot study of psilocybin enhanced group psychotherapy in patients with cancer. *J Pain Symptom Manage* 2023;66:258-69. doi:10.1016/j.jpainsymman.2023.06.006
172 Murphy R, Kettner H, Zeifman R, et al. Therapeutic alliance and rapport modulate responses to psilocybin assisted therapy for depression. *Front Pharmacol* 2022;12:788155. doi:10.3389/fphar.2021.788155
173 Levin AW, Lancelotta R, Sepeda ND, et al. The therapeutic alliance between study participants and intervention facilitators is associated with acute effects and clinical outcomes in a psilocybin-assisted therapy trial for major depressive disorder. *PLoS One* 2024;19:e0300501. doi:10.1371/journal.pone.0300501
174 Aday JS, McAfee J, Conroy DA, et al. Preliminary safety and effectiveness of psilocybin-assisted therapy in adults with fibromyalgia: an open-label pilot clinical trial. *Front Pain Res (Lausanne)* 2025;6:1527783. doi:10.3389/fpain.2025.1527783
175 Schindler EAD, Sewell RA, Gottschalk CH, et al. Psilocybin pulse regimen reduces cluster headache attack frequency in the blinded extension phase of a randomized controlled trial. *J Neurol Sci* 2024;460:122993. doi:10.1016/j.jns.2024.122993
176 Bradley ER, Sakai K, Fernandes-Osterhold G, et al. Psilocybin therapy for mood dysfunction in Parkinson's disease: an open-label pilot trial. *Neuropsychopharmacology* 2025;50:1200-9. doi:10.1038/s41386-025-02097-0
177 Michaels TI, Purdon J, Collins A, Williams MT. Inclusion of people of color in psychedelic-assisted psychotherapy: a review of the literature. *BMC Psychiatry* 2018;18:245. doi:10.1186/s12888-018-1824-6
178 Hughes ME, Garcia-Romeu A. Ethnoracial inclusion in clinical trials of psychedelics: a systematic review. *EClinicalMedicine* 2024;74:102711. doi:10.1016/j.eclinm.2024.102711
179 Grossman DH, Madden KR, Mehtani NJ, et al. A systematic review of income and education reporting in psychedelic clinical trials. *Nat Ment Health* 2025;3:567-74. doi:10.1038/s44220-025-00417-3
180 Fogg C, Michaels TI, de la Salle S, Jahn ZW, Williams MT. Ethnoracial health disparities and the ethnopsychopharmacology of psychedelic-assisted psychotherapies. *Exp Clin Psychopharmacol* 2021;29:539-54. doi:10.1037/pha0000490
181 Marseille E, Mitchell JM, Kahn JG. Updated cost-effectiveness of MDMA-assisted therapy for the treatment of posttraumatic stress disorder in the United States: findings from a phase 3 trial. *PLoS One* 2022;17:e0263252. doi:10.1371/journal.pone.0263252
182 Marseille E, Stauffer CS, Agrawal M, et al. Group psychedelic therapy: empirical estimates of cost-savings and improved access. *Front Psychiatry* 2023;14:1293243. doi:10.3389/fpsyt.2023.1293243
183 Mithoefer MC. A manual for MDMA-assisted psychotherapy in the treatment of posttraumatic stress disorder. Multidisciplinary Association for Psychedelic Studies. 2017 (cited 17 Nov 2024). Available from: https://maps.org/wp-content/uploads/2022/05/MDMA-Assisted-Psychotherapy-Treatment-Manual-V8.1-22AUG2017.pdf
184 Palitsky R, Maples-Keller JL, Peacock C, et al. A critical evaluation of psilocybin-assisted therapy protocol components from clinical trial patients, facilitators, and caregivers. *Psychotherapy (Chic)* 2025;62:348-62. doi:10.1037/pst0000551

Web appendix: Supplement

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